



SACS

Gap

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Gap program module allows for the analysis of nonlinear one-way elements. Gap uses linear combinations of gap element unit axial load cases to eliminate loading in any gap element that contradicts the element type designation (i.e. tension in a compression only element, compression in a tension only element, etc.).

1.2 PROGRAM FEATURES

Gap requires a SACS input model file and an optional Gap input file for execution. The program coupled with SACS IV and Combine creates a common solution file containing results of the linear combination of the designated load combinations and gap element unit axial load cases.

Some of the main features and capabilities of the Gap program module are:

1. Allows simulation of compression only, tension only and no load elements.
2. Simulation of nonlinear springs defined by specified force/deflection curves.
3. Multiple load combinations may be analyzed in a single execution.
4. Ability to simulate loss of support and/or check differential settling problems when used in conjunction with specified displacements.
5. Ability to simulate tension or compression only supports.
6. Ability to define load combinations in the Gap input file.
7. Ability to override gap element type for particular load cases.

1.3 PROGRAM STRUCTURE

When executing a Gap analysis, a static analysis is performed for output load cases. For each output load case, the following procedure is used to obtain gap analysis results:

1. The load in each gap element is checked. Any gap element having loading that contradicts the element type specified (i.e. compression only element with tension load) is flagged.
2. A dummy unit load case corresponding to the gap element with the highest contradicting load is added to the combination then factored until the loading in that gap element is negated.
3. The load in each gap element is rechecked. If any gap element contradicts the type specified, the procedure is repeated.
4. The output load case then consists of the original basic load conditions and any gap dummy unit load cases factored to eliminate any contradicting gap element loading.

2 GAP MODELING AND INPUT

The Gap program requires a SACS model file. A Gap input file is optional and is not required for most gap analyses.

The model file may require some minor modeling consideration for the purpose of gap analysis. Gap element data for 'no load', 'compression only' and 'tension only' elements may be specified directly in the model file or can be optionally defined outside of the model in a Gap input file. Gap element data for 'force deflection' elements, on the other hand, must be specified in a Gap input file. 'Friction' element data can be specified only in the model file.

2.1 DEFINING GAP ELEMENT DATA DIRECTLY IN THE MODEL

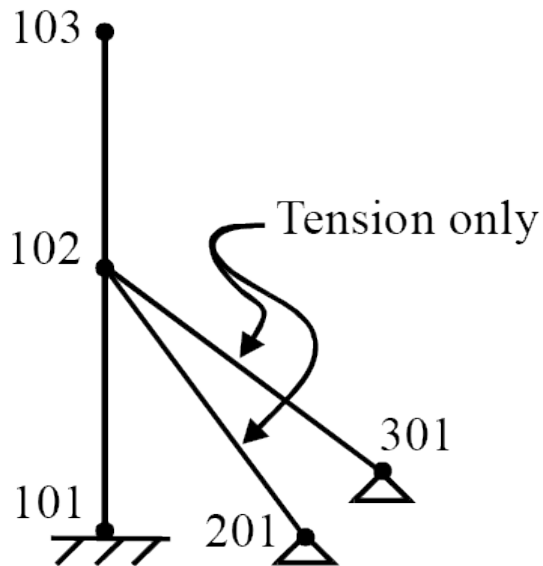
Any element in the model may be regarded as a gap one-way element. Some minor modeling considerations may be required for execution of a gap analysis.

2.1.1 Defining Gap Elements

Elements may be designated as one of the following nonlinear gap element types; 'compression only', 'tension only', 'no load' or 'friction' element. Elements that are to be considered as one of these types of nonlinear gap elements are designated in the model file as part of the member group data on the GRUP line or member data on the MEMBER line.

Note: Compression only, tension only, and no-load gap elements may optionally be designated in a Gap input file. Force-deflection gap elements must be designated in the Gap input file. Friction elements must be designated in the model file, and different friction coefficients in two directions are defined in SACS Executive Solver Option.

When designating an entire member group as a gap element, the element type is designated in column 30 of the GRUP line by 'C' (compression only), 'T' (tension only) or 'N' (no load). Individual members may be defined as gap elements by specifying the element type in column 22 on the MEMBER line as 'C' (compression only), 'T' (tension only) or 'N' (no load).



The following defines members 102-201 and 102-301 as tension only gap elements.

	1	2	3	4	5	6	7	8
1	MEMBER	102	201	GAP	T011			
2	MEMBER	102	301	GAP	T011			

2.1.2 Member End Releases

Member end releases may be used to remove shear, bending and/or torsion capacity of elements that will be declared as gap elements.

For example, a guyed structure utilizing tension only gap elements may require member end releases as follows:

	1	2	3	4	5	6	7	8
1	MEMBER	102	201	GAP	T011			
2	MEMBER	102	301	GAP	T011			

Note: The axial degree of freedom at the ends of an element designated as a compression or tension only element should not be released.

2.1.3 Designating Load Cases to Analyze

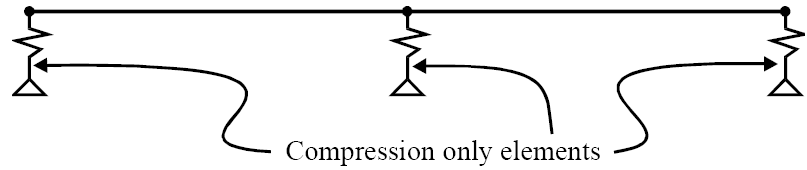
Load cases that are to be analyzed as part of the gap element analysis are designated on the LCSEL line in the model file or in the Seastate input file using the 'GP', 'ST' or default (ie. blank) option.

Note: If no LCSEL line with the 'GP' option is specified, then load cases designated for a static analysis (i.e. 'ST' or blank option) are used.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LCSEL	A1	A2	A3	A7			

2.1.4 Simulating Compression or Tension Only Supports

Gap elements may be used to simulate compression only or tension only supports by modeling two joints at the support. One joint is designated as the support joint and assigned the appropriate fixities, while the other joint is attached to the structure. A gap element is then modeled between the two joints. See the figure below.



Note: The gap element must be modeled such that the length is greater than zero. This can be done by either separating the joints at the support or by defining a member offset.

2.1.5 Overriding Gap Type for a Load Case

The gap element type may be changed from load case to load case. This requires a Gap input file including a GAPOPT line and the LCGAP line.

Gap type may be overridden for individual members or for groups of members using the LCGAP with 'MEM' or 'GRP' specified in columns 19-21, respectively.

Enter the load case to which the override applies in columns 7-10 and the type of gap element in columns 16-17. Specify 'CO', 'TO' or 'NL' for compression only, tension only or no-load elements, respectively. Enter 'INC' in columns 12-14 if only the members or groups specified in columns 23-77 are to be changed as designated, or 'EXC' if all gap elements except the members or groups specified in columns 23-77 are to be changed.

Note: Because gap elements may be either included or excluded for a particular load case, the 'INC' and 'EXC' features are mutually exclusive and cannot be mixed within a particular load case.

The following sample designates that members 1101-2101,1103-2103 and all gap elements in member group 'CAN' are to be considered as no load elements for load case 'CMB1':

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPOPT	4	EN					
2	LCGAP	CMB1	INC	NL	MEM	11012101	11032103	
3	LCGAP	CMB1	INC	NL	GRP	CAN		
4	END							

Note: The LCGAP feature is only applicable for gap elements defined in the model file.

2.1.6 Dummy Unit Load Case

Dummy unit load cases are not required for gap elements defined in the model file.

2.1.7 Gap Advanced Solver Options

SACS models with a large number of gap elements (like Suction Buckets with nonlinear springs) require a more efficient numerical solver to speed up the Gap analysis. The Gap program provides new options to reduce the run-time of a Gap analysis for such models. The Advanced Numerical Solver options can be chosen on columns 77-78 of the GAPOPT input line. Two new options are available:

- **Advanced-1: Efficient Matrix Inversion:** This is similar to the Gap standard iterative solver but the matrix inversion calculation is replaced by an efficient matrix inversion algorithm. See the Commentary Section for additional details
 - This option can be selected by entering 'A1' on columns 77-78 of the GAPOPT line
- **Advanced-2: Efficient Linear System Solve:** Using this option, the Gap iterative solver does not require a matrix inversion process and uses an efficient matrix factorization algorithm. See the Commentary Section for additional details
 - This option can be selected by entering 'A2' on columns 77-78 of the GAPOPT line
- **Direct Solver:** If the model has *only* no-load, tension-only, and/or compression-only gap elements with *zero initial gap*, the gap analysis can be performed using a direct solver instead of the default iterative solver.
 - This option can be selected by entering 'AD' on columns 77-78 of the GAPOPT line

Note: If the gap program cannot use the direct solver for any reason, it automatically switches to Advanced-2: Efficient Linear System Solve ('A2') option.

The default value is a standard iterative Gap solver without a high-performance algorithm.

Note: For a model with a small number of gap elements, the speed-up gain from either of the above options is minimal. However, for a jacket with thousands of gap elements, using one of the above options is highly recommended to speed up the analysis.

2.1.8 Gap Continue Option

By default, the Gap program terminates the analysis with an error message if a given load condition does not converge within the specified number of iterations to the convergence tolerance entered on the GAPOPT line. However, under some conditions, it is desirable to continue the analysis even though convergence has not been achieved (for example, when the residual is relatively small but is not smaller than the specified tolerance).

The default non-convergence criterion can be overridden by entering 'NE' on columns 79-80 of the GAPOPT line allowing the Gap analysis to continue with an output warning message indicating non-convergence of load cases.

Note: In the case of the non-convergence, the model response (displacement and internal forces) must be carefully checked to ensure the non-convergence does not lead to significant error in the analysis results.

2.2 DEFINING GAP ELEMENT DATA OUTSIDE OF THE MODEL

Most Gap analyses require no special input other than defining the gap elements and designated load cases to analyze in the model file. Using force-deflection elements, however, requires using a Gap input file.

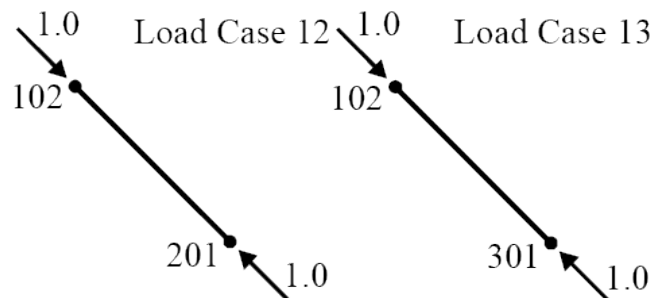
Note: When defining any force deflection gap element data in a Gap input file, all other gap element data must also be defined in the Gap input file also.

2.2.1 Modeling Considerations

Certain modeling considerations must be addressed when defining gap element data in a Gap input file.

2.2.1.1 Defining Dummy Unit Load Cases

Any gap elements defined in the Gap input file require a dummy load case in the model. For each gap element defined, a dummy load case consisting of only a unit axial load for that gap element must be specified by the user.



When creating a dummy load case for a gap element, the net load of the dummy unit load case must be zero. For example, load cases 12 and 13 define a unit load axial load for members 102-201 and 102-301 from the preceding figure, respectively.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOADCN	12						
2	LOAD X	102 201	0.0	1.0		MEMB CONC	DUMMY12	
3	LOAD X	201 102	0.0	-1.0		MEMB CONC	DUMMY12	
4	LOADCN	13						
5	LOAD X	102 301	0.0	1.0		MEMB CONC	DUMMY13	
6	LOAD X	301 102	0.0	-1.0		MEMB CONC	DUMMY13	

Note: Either a tension or compression unit axial load may be defined. Dummy load cases should produce no net load on the structure.

The procedure illustrated above uses two 'LOAD' input lines per dummy unit load case. The first load is defined as a compressive axial load at the begin joint while the second load is defined as a compressive load at the end joint.

2.2.2 Gap Input File Requirements

When using a Gap input file to specify gap element data, all input data including input and output load cases data must be specified in the Gap input file. The following sections address the specification of this input data.

2.2.2.1 Analysis Units

Gap analysis units are designated on the GAPOPT input line in columns 21-22.

2.2.2.2 Designating Basic Load Cases

Basic load cases in the model input file that are referenced by a load combination defined in the Gap input file are called 'Real' load cases and must be designated as such. The total number of 'Real' load cases referenced must be specified in columns 7-10 on the GAPOPT input line. Each 'Real' load case must also be listed on the LCSEL input line in the Gap input file. For example, load cases A1, A2, A3 and A7 may be defined as 'Real' basic load cases as follows:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPOPT	4	EN					
2	LCSEL		A1	A2	A3	A7		

2.2.2.3 Designating Load Cases to Analyze

The Gap program analyzes load combinations and basic load cases designated as 'Output' load cases in the Gap input file.

In general, 'Output' load cases are load combinations consisting of 'Real' load cases that are defined in the model file and designated as such in the Gap input file. The number of 'Output' load cases must be specified in columns 11-14 on the GAPOPT line.

2.2.2.4 Defining Output Load Cases

An 'Output' load case or combination may consist of up to 48 'Real' load cases designated on the LCSEL and is defined in the Gap input file using LCOMB input lines.

The load combination name is specified in columns 7-10. For each 'Real' load case, the load factor to be applied and the load case name are specified. The allowable stress modifier for the load case is specified in columns 72-76.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LCSEL		A1	A2	A3	A7		
2	LCOMB	CMB1	A1	1.0	A2	0.75		
3	LCOMB	CMB2	A1	1.0	A2	3.00		1.33

2.2.2.5 Designating Gap Elements

Elements may be designated as one of the following nonlinear gap element types; compression only, tension only, no load, or force-deflection element. Elements that are to be considered nonlinear gap elements are designated in the Gap input file using the GAPELM line.

The element connecting joints are specified in columns 7-11 and 12-16. The element type is designated in columns 24-25 by 'CO' (compression only), 'TO' (tension only), 'NL' (no load), 'FD' (force-deflection) or 'RP' (repeat previous element type). The dummy unit load case defined in the model file that corresponds to the element being defined is specified in columns 17-21.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPELM	101	201	4	CO			
2	GAPELM	103	203	5	NL			

Note: Dummy load cases for any gap element defined using the GAPELM line must be added to the model by the user manually. The dummy load case corresponding to the element should contain only unit axial load at each end of the element and should add no net load to the overall structure.

2.2.2.6 Defining Force-Deflection Curve

The curve used for an element designated as force-deflection type gap element is defined on the line labeled F-DEL immediately following the GAPELM line used to designate the element as a force-deflection gap element.

Points on the curve are entered in columns 9-80 and must be entered in order of increasing deflection. As many F-DEL lines as needed to define the curve may be used.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPELM	101	201	4	CO			
2	GAPELM	103	203	5	FD			
3	F-DEL		0.0	-10.0	0.0	0.0	100.0	1.0
4	F-DEL		300.0	10.0				2.0
5	GAPELM	105	205	6	CO			

Note: For deflections outside of the range of the curve, the force value is assumed to be constant.

To plot the force/deflection curve in a force-deflection type gap element, enter either 'PFD' or 'PFG' in columns 45-47 of the GAPOPT line in the Gap input file. The 'PFD' option produces a force/deflection curve without grid lines; the 'PFG' option produces the same with grid lines.

2.2.2.7 Defining Friction Data

Friction elements that are defined in the GAP input file are no longer supported. Friction elements data must be specified in the model file.

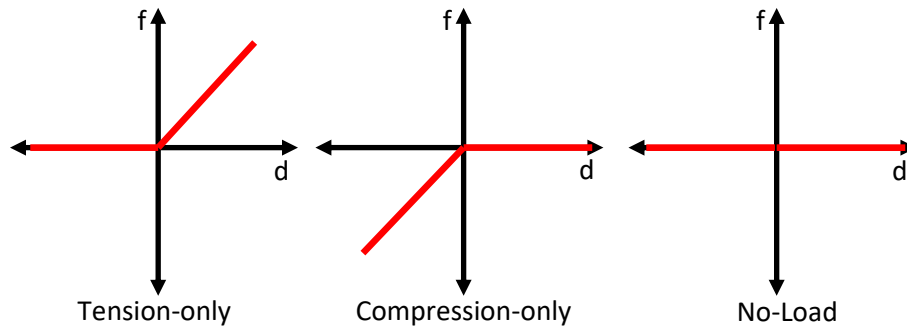
3 COMMENTARY

3.1 GAP ELEMENT FORCES

To solve various gap element types (tension-only, compression-only, force-deflection, and friction), the Gap program converts all gap elements to the force-deflection element as follows:

- **Tension-only:** a force-deflection element with zero force in compression, linear force-deflection in the tension. The program uses member axial stiffness in the tension
- **Compression-only:** a force-deflection element with zero force in tension, linear force-deflection in the compression. The program uses member axial stiffness in the compression
- **No Load:** zero loads and zero stiffness force-deflection
- **Force-deflection:** No change, the program uses the user-defined force-deflection curves
- **Friction:** The program uses three gap elements: one compression-only element in local X (to model contact) and two friction elements in the Local Y and Z. The load in the friction element is set by the friction coefficient and the axial load in the local X compression-only gap element.

The following figure illustrates the force-deflection curves for different element types:



The Gap program module uses the relationship between force and load factors of elements designated as nonlinear gap elements. For any output load condition, the total force F_i in a gap element i , can be expressed in terms of the initial gap element force, F_{o_i} , and the force/load factor relationship of the element as follows

$$F_i = F_{o_i} + \sum_{j=1}^n (A_{ij}P_j) \quad (1)$$

where n is the number of gap elements, A_{ij} is the force in gap element i due to a unit load factor, P_j , applied to the j^{th} dummy load case. The j^{th} dummy load case corresponds to a unit load case applied to the j^{th} gap element. The above equation can be rewritten in a matrix form as:

$$\{F\} = \{F_o\} + [A]\{P\} \quad (2)$$

Depending on the gap element types, the above equation may be solved *directly* or *iteratively*, as presented below

1. **Direct Solver:** If a SACS model *only* has tension-only, compression-only, and/or no-load elements (not other gap element types), and the *initial gap values* for all elements are zero, the load factors can be determined by solving equation (2) *directly*.
The direct solver is available by entering option 'AD' on the GAPOPT input line. For this option, the Gap program first checks the model to ensure the model has only compatible elements (tension-only, compression-only, and/or no-load), and all initial gap values are zero. If the program cannot use the direct solver for any reason, it automatically switches to the iterative solver with the *Advanced Matrix Solver* ('A2' option).
2. **Iterative Solver:** if a SACS model has friction elements or force-deflection elements in addition to tension-only, compression-only, and/or no-load elements, or the initial gap values are not zero, the equation (2) must be solved iterative using Newton-Raphson algorithm – as discussed in the next section.

3.2 Iterative Solver

In the case of the iterative solver, the load factors $\{P\}$ must be determined such that the difference between element force $\{F\}$ and the force computed based on the force-deflection curves $\{F_s\}$ must be zero. In the other words:

$$\{R\} = \{F\} - \{F_s\} = 0 \quad (3)$$

The above equation is nonlinear and can be solved using the Newton-Raphson method. Thus, the first step is to compute the derivative of $\{R\}$ (residual) with respect to the load factor $\{P\}$.

$$\frac{d\{R\}}{d\{P\}} = \frac{d\{F\}}{d\{P\}} - \frac{d\{F_s\}}{d\{P\}} \quad (4)$$

The first term $\frac{d\{F\}}{d\{P\}}$ is readily available based on Equation (2) where $\{F_o\}$ is constant:

$$\frac{d\{F\}}{d\{P\}} = [A] \quad (5)$$

The second term can be determined as:

$$\frac{dF_{si}}{dP_j} = \frac{dF_{si}}{du_i} \frac{du_i}{dP_j} \quad (6)$$

in which u_i is the displacement vector for the gap elements. The first term $S_i = \frac{dF_{si}}{du_i}$ is simply the slope of the force-deflection curve. To compute the second term, the displacement of i^{th} gap element is first written as a function of the gap force F_i and load factor P_i as:

$$u_i = \frac{F_i - P_i f_{idummy}}{K_i} \quad (7)$$

where K_i is the element stiffness (e.g. axial stiffness) and f_{idummy} is the applied dummy load at i^{th} gap element. It is worth noting that F_i is also a function of load factor $\{P\}$, and its derivative with respect to the load factor P_i must be taken to account. Therefore:

$$\frac{du_i}{dP_j} = \begin{cases} i = j & \frac{A_{ii} - f_{idummy}}{K_i} \\ i \neq j & \frac{A_{ij}}{K_i} \end{cases} \quad (8)$$

Substituting Equation (8) into Equation (6):

$$\bar{A}_{ij} = \frac{dR_i}{dP_j} = \begin{cases} i = j & A_{ii} - \left(\frac{A_{ii} - f_{idummy}}{K_i} \right) S_i \\ i \neq j & A_{ij} - \left(\frac{A_{ij}}{K_i} \right) S_i \end{cases} \quad (9)$$

Finally, following the Newton-Raphson iteration, the load factor increment can be determined as

$$\Delta\{P\} = -[\bar{A}]^{-1}(\{F\} - \{F_s\}) \quad (10)$$

Note: it can be concluded that the system does not have a unique solution when the determinant of $\bar{A}_{ij}$ is zero. Moreover, the instability of the system increases when the determinant approaches zero. $\bar{A}_{ij}$ is assumed to be a symmetric positive definite matrix, so diagonal components affect the determinant. The program returns a warning message showing the element which causes the determinant to get closer to zero.

The Gap program iterative solver can be summarized in the following steps

1. Determine force-deflection curves for all gap elements
2. **Initialize:** $\{P\} = 0$
3. **Calculate the gap elements forces:** $\{F\} = \{F_o\} + [A]\{P\}$
4. **Calculate gap element displacement:** $u_i = \frac{F_i - P_i f_{idummy}}{K_i}$
5. **Update the force F_{s_i} and slope S_i :** Using the force-deflection curves and displacements
6. **Calculate residual:** $\{R\} = \{F\} - \{F_s\}$
7. **Check for the convergence:** if the maximum error in $\{R\}$ is less than the tolerance stop the iteration
8. **Determine the load factor increment:** $\Delta\{P\} = -[\bar{A}]^{-1}(\{F\} - \{F_s\})$
9. **Update the load factor:** $\{P\}_{new} = \{P\}_{previous} + \Delta\{P\}$
Go to step 3 and repeat until convergence achieves

3.2.1 Advanced Iterative Solvers

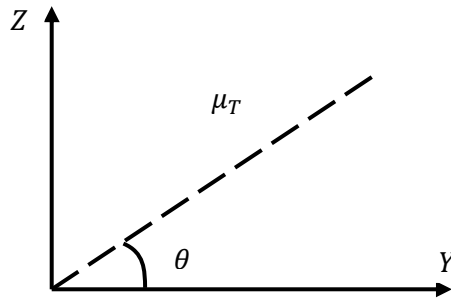
As illustrated in the previous section, the Gap iterative solver relies on inverting the matrix $[\bar{A}]$ in Step 8 of the algorithm. When a model has only a few numbers of the gap elements, the $[\bar{A}]^{-1}$ can be computed efficiently with minimal effort. However, when there are thousands of gap elements (for example in the case of suction bucket analysis), the computational cost associated with $[\bar{A}]^{-1}$ greatly increases and the Gap program slows down. To speed up the Gap analysis, two new advanced options have been introduced to the Gap program.

- **Advanced Matrix Inversion:** This option utilizes an efficient numerical algorithm to compute $[\bar{A}]^{-1}$ in step 8 of the above algorithm.
- **Advanced Matrix Solver:** This option refactors Equation (10) as a linear system of equations $[\bar{A}]\Delta\{P\} = -(\{F\} - \{F_s\})$, and it utilizes an efficient numerical solver to solve this linear system instead of inverting the matrix.

3.3 DIFFERENT FRICTION COEFFICIENT IN TWO DIRECTIONS TREATMENT

SACS treats friction elements with three different distinguished components in local coordinates. Local X coordinate shows the normal force and forces along Y and Z coordinates show sliding force (friction force components). For some specific structures or components, the friction coefficient might be different in two directions. SACS uses the following method to find the solution for a friction element with different friction coefficients through an iteration process.

Let's assume that friction coefficient in Y direction is μ_y and friction coefficient in Z direction is μ_z and $\mu_y \neq \mu_z$. It is obvious that for the case that $\mu_y = \mu_z$, friction coefficient in any desired direction (μ_T) would be $\mu_T = \mu_y = \mu_z$. To find the friction coefficient (μ_T) in the direction with an angle of θ with Y direction (following figure), it has been assumed that an "Ellipse" is passing through the value of friction coefficient along each axis.



Therefore, μ_T is located in an Ellipse created by μ_y and μ_z and can be calculated using the following relation:

$$\mu_T = \frac{\mu_y \mu_z}{\sqrt{\mu_y^2 \sin^2 \theta + \mu_z^2 \cos^2 \theta}}$$

With this approach, μ_T will be equal to μ_y and μ_z when θ is equal to 0 and $\frac{\pi}{2}$ respectively. The program will iterate to find components (i.e. F_y and F_z) in a way that, if one finds the angle θ and calculates μ_T using the previous formula, the resultant force will follow the relation given as:

$$\sqrt{F_y^2 + F_z^2} \leq \mu_T F_x$$

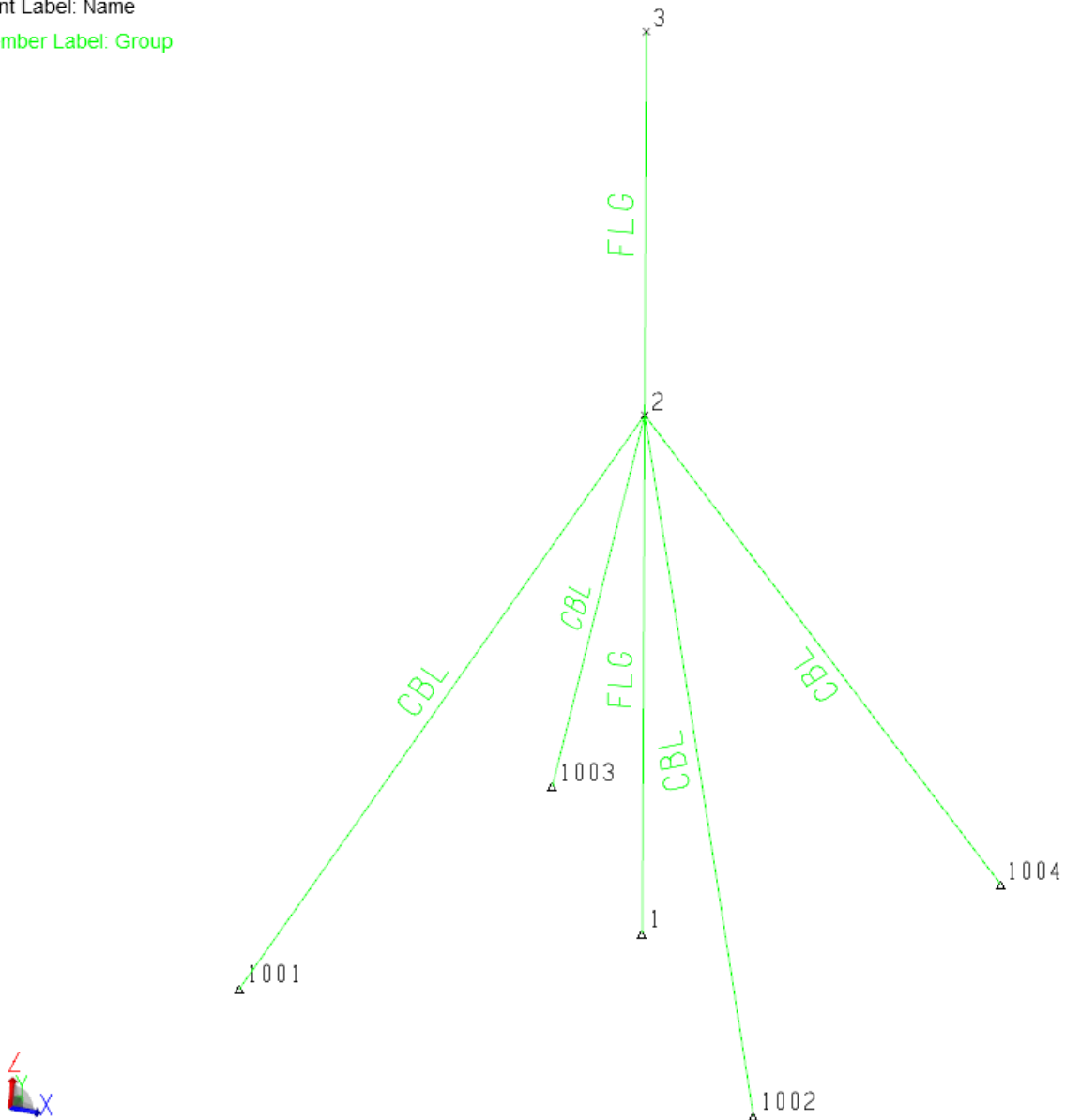
4 SAMPLE PROBLEMS

4.1 SAMPLE PROBLEM 1 – TOWER STRUCTURE WITH TENSION ONLY GAP ELEMENTS

Sample Problem 1 is a tower structure supported laterally by four wire cables as shown below. The wire cables, group 'CBL', are simulated using tension only gap elements with shear capacity member releases at joint 2.

Joint Label: Name

Member Label: Group



Two basic load conditions are specified; load case WGT structure self-weight and load case LAT a lateral load in the positive global X direction applied at joint 3.

No Gap input file was required for this analysis. All required gap analysis data information was specified in the model file that follows:

```

1      1234567890123456789012345678901234567890123456789012345678901234567890
2      OPTIONS      EN      SDUC      1 1 DDC C      PT PTPT      PT
3      LCSEL GP      CMB1 CMB2
4      SECT
5      SECT CABLE      TUB 3.75      0.10      0.10      0.10      0.010.001
6      GRUP
7      GRUP CBL CABLE      13.0011.6036.00 1      1.001.00      0.50N490.00
8      GRUP FLG      24.00 0.625      29.0011.6036.00 1      1.001.00      0.50N490.00
9      MEMBER
10     MEMBER      1 2 FLG
11     MEMBER      2 3 FLG
12     MEMBER 1001 2 CBL T      011
13     MEMBER 1002 2 CBL T      011
14     MEMBER 1003 2 CBL T      011
15     MEMBER 1004 2 CBL T      011
16     JOINT
17     JOINT      1      111111
18     JOINT      2      30.000
19     JOINT      3      50.000
20     JOINT 1001 -15.000-15.000      111000
21     JOINT 1002 15.000-15.000      111000
22     JOINT 1003 -15.000 15.000      111000
23     JOINT 1004 15.000 15.000      111000
24     LOAD
25     LOADCN WGT
26     LOAD Z      1 2      -0.033      -0.033      GLOB UNIF      DEADWT
27     LOAD Z      2 3      -0.033      -0.033      GLOB UNIF      DEADWT
28     LOADCN LAT
29     LOAD Z      3      25.0000      GLOB JOIN      LATERAL
30     LOAD Z      2      25.0000      GLOB JOIN      LATERAL
31     LCOMB
32     LCOMB CMB1 WGT 1.0 LAT 1.0
33     LCOMB CMB2 WGT 1.0 LAT -1.0
34     END

```

The structure was analyzed for the two load combinations of the self-weight and the lateral load defined as CMB1 and CMB2. Below is a description of the pertinent input data:

2 Load combinations CMB1 and CMB2 are to be analyzed as designated on the LCSEL line.

11 - 14 Members 1001-2, 1002-2, 1003-2 and 1004-2 are defined as tension only gap elements by 'T' in column 22.

31 Load combination CMB1 contains 100% of load cases WGT and LAT.

32 Load combination CMB2 contains 100% of load cases WGT and LAT but the direction of load case LAT is switched to the negative X direction by specifying a factor of -1.

The following is a portion of the output listing file created by the GAP program module:

PRE VERSION 12.1.0.14

** PROGRAM OPTIONS **

UNITS

....ENGLISH

EXECUTION

....GAP ANALYSIS
....SHEAR DEFORMATION INCLUDED
....UNITY CHECK
WSD AISC 9TH EDITION WITH API-RP2A 21ST EDITION FOR TUBULARS
....PLATE PANEL UNITY CHECK (DEFAULT)
DNV-RP-C201
....DKT PLATES SELECTED
....NO SEGMENTS FOR PRISMATIC MEMBERS 1
....NO SEGMENTS/SECTION FOR NON-PRISMATIC MEMBERS 1

REPORTS SELECTED

....JOINT DEFLECTIONS.....PRINT
....ELEMENT STRESS AT MAXIMUM UNITY CHECK...PRINT
....BEAM COMBINED AND SHEAR UNITY CHECK.....PRINT
....ELEMENT INTERNAL LOADS.....PRINT

LOAD

....NO. BASIC LOAD COND. 2
....NO. COMB. LOAD COND. 2

TUBULAR MEMBER PROPERTIES

GRP	M/S	JOINT THICK FT	WALL THICK IN	OUTSIDE DIAM. IN	E 1000 KSI	G 1000 KSI	AXIAL AREA IN**2	***** X-X IN**4	MOMENTS OF Y-Y IN**4	INERTIA Z-Z IN**4	***** YIELD STRESS KSI	KY	KZ	SHEAR AREA IN**2	RING SPACE FT	SECT LENG FT	TAPER
CBL	1	0.00	0.001	0.01	13.0	11.6	3.7500	0.10000	0.10000	0.10000	36.0	1.0	1.0	1.88	0.00	0.00	
FLG	1	0.00	0.625	24.00	29.0	11.6	45.897	6273.9	3136.9	3136.9	36.0	1.0	1.0	22.95	0.00	0.00	

OPTIMIZATION DATA

OPTIMIZED FINAL BANDWIDTH = 1.857

FINAL MAXIMUM BANDWIDTH = 5

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:45 PRE PAGE 4

**** LOAD CASE STATUS REPORT ****

LOAD CASE	LOAD ID	PRINT OPTION	DEAD LOAD	P-DELTA LOAD	LOAD FACTOR	AMOD FACTOR
1	WGT	NO	NO	NO	1.00	1.00
2	LAT	NO	NO	NO	1.00	1.00
3	G001	NO	NO	NO	1.00	1.00
4	G002	NO	NO	NO	1.00	1.00
5	G003	NO	NO	NO	1.00	1.00
6	G004	NO	NO	NO	1.00	1.00
7	CMB1	YES	NO	NO	1.00	1.00
8	CMB2	YES	NO	NO	1.00	1.00

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:46 SLV PAGE 1

SLV VERSION 12.1.0.13

** SACS PROBLEM DESCRIPTION **

NUMBER OF JOINTS	7
NUMBER OF BEAMS	6
NUMBER OF PLATES	0
NUMBER OF SHELLS	0
NUMBER OF SOLIDS	0
NUMBER OF LOADS	6
NUMBER OF MASTER JOINTS ...	0
PRINT OPTION	0

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:46 SLV PAGE 2

APPLIED LOAD SUMMARY

LOAD CASE NO. ID	TOTAL FORCE (X) KIPS	TOTAL FORCE (Y) KIPS	TOTAL FORCE (Z) KIPS
1 WGT	0.000000E+00	0.000000E+00	-1.649999E+00
2 LAT	5.000000E+01	0.000000E+00	0.000000E+00
3 G001	0.000000E+00	0.000000E+00	0.000000E+00
4 G002	0.000000E+00	0.000000E+00	0.000000E+00
5 G003	0.000000E+00	0.000000E+00	0.000000E+00
6 G004	0.000000E+00	0.000000E+00	0.000000E+00

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:46 GAP PAGE 3

***** GAP ELEMENT REPORT *****

GAP ELEMENT	1ST JOINT	2ND JOINT	GRUP ID	LOAD CASE	ELEMENT TYPE	DEGREE OF FREEDOM	PRESET GAP IN	**** FRICTION ELEMENT **** COEFF. FRICT.	NORMAL J1	MEMBER J2	LOAD TYPE	1ST POINT FORCE KIPS	DEFL. IN	2ND POINT FORCE KIPS	DEFL. IN
1	1001	2	CBL	G001	TENS	AXIAL	0.00								
2	1002	2	CBL	G002	TENS	AXIAL	0.00								
3	1003	2	CBL	G003	TENS	AXIAL	0.00								
4	1004	2	CBL	G004	TENS	AXIAL	0.00								

LOAD CASE CMB1 CONVERGED IN 30 ITERATIONS

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:46 GAP PAGE 4

NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	1001-	2	TENS	0.71197	78.7
2	1002-	2	TENS	-0.76926	0.0
3	1003-	2	TENS	0.71197	78.7
4	1004-	2	TENS	-0.76926	0.0

LOAD CASE CMB2 CONVERGED IN 30 ITERATIONS

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:46 GAP PAGE 5

NO.	GAP ELEMENT		TYPE	DEFLECTION IN	FORCE KIPS	FACTOR						
1	1001-	2	TENS	-0.76926	0.0	0.850543E+02						
2	1002-	2	TENS	0.71197	78.7	0.000000E+00						
3	1003-	2	TENS	-0.76926	0.0	0.850543E+02						
4	1004-	2	TENS	0.71197	78.7	0.000000E+00						
SACS CONNECT Edition (v11.0)				Bentley Systems			ID=Ym5ng2NpaiJxrIF8nIen					
							DATE 15-AUG-2018	TIME 15:19:47	CMB PAGE	1		
							SCB VERSION 12.1.0.14					
				NUMBER OF FINAL LOAD CASES			2					
LOAD CASE	NUMB COMB	COMB TYPE	AMOD FACT	LOAD CASE DESCRIPTION			UNIT ORIG LC =	FACTOR	SIGN ROT XYZ	CHANGE DEFL XYZ	STRES	
CMB1	4	1.000					P WGT	1.000	000	000	0	
							P LAT	1.000	000	000	0	
							P G002	85.054	000	000	0	
							P G004	85.054	000	000	0	
CMB2	4	1.000					P WGT	1.000	000	000	0	
							P LAT	-1.000	000	000	0	
							P G001	85.054	000	000	0	
							P G003	85.054	000	000	0	

SACS CONNECT Edition (v11.0)				Bentley Systems				ID=Ym5ng2NpaIJxrIF8nIen			
								DATE 15-AUG-2018	TIME 15:19:47	PST PAGE	6
SACS-IV SYSTEM MEMBER FORCES AND MOMENTS											
***** KIPS *****											
***** IN-KIPS *****											
MEMBER	MEMBER	GROUP	LOAD	FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)		
NUMBER	END	ID	CASE								
1-	2	1	FLG CMB1	-130.20	14.28	0.00	0.00	0.00	860.86		
			CMB2	-130.20	-14.28	0.00	0.00	0.00	-860.86		
	2		CMB1	-129.21	14.28	0.00	0.00	0.00	6000.00		

				CMB2	-129.21	-14.28	0.00	0.00	0.00	-6000.00
2-	3	2	FLG	CMB1	-0.66	-25.00	0.00	0.00	0.00	6000.00
				CMB2	-0.66	25.00	0.00	0.00	0.00	-6000.00
		3		CMB1	0.00	-25.00	0.00	0.00	0.00	0.00
				CMB2	0.00	25.00	0.00	0.00	0.00	0.00
1001-	2	1001	CBL	CMB1	78.72	0.00	0.00	0.00	0.00	0.00
				CMB2	0.00	0.00	0.00	0.00	0.00	0.00
		2		CMB1	78.72	0.00	0.00	0.00	0.00	0.00
				CMB2	0.00	0.00	0.00	0.00	0.00	0.00
1002-	2	1002	CBL	CMB1	0.00	0.00	0.00	0.00	0.00	0.00
				CMB2	78.72	0.00	0.00	0.00	0.00	0.00
		2		CMB1	0.00	0.00	0.00	0.00	0.00	0.00
				CMB2	78.72	0.00	0.00	0.00	0.00	0.00
1003-	2	1003	CBL	CMB1	78.72	0.00	0.00	0.00	0.00	0.00
				CMB2	0.00	0.00	0.00	0.00	0.00	0.00
		2		CMB1	78.72	0.00	0.00	0.00	0.00	0.00
				CMB2	0.00	0.00	0.00	0.00	0.00	0.00
1004-	2	1004	CBL	CMB1	0.00	0.00	0.00	0.00	0.00	0.00
				CMB2	78.72	0.00	0.00	0.00	0.00	0.00
		2		CMB1	0.00	0.00	0.00	0.00	0.00	0.00
				CMB2	78.72	0.00	0.00	0.00	0.00	0.00

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:19:47 PST PAGE 7

SACS-IV SYSTEM ELEMENT STRESS REPORT AT MAXIMUM UNITY CHECK

MEMBER	GRP	MAXIMUM UNITY CHECK	CRITICAL COND.	LOAD CASE NO.	DIST FROM END FT	***** APPLIED STRESSES *****					* CM VALUES *		* NEXT TWO HIGHEST CASES *				
						AXIAL	** BENDING **		*** SHEAR ***	Y	Z	Y	Z	UNITY CHECK	LOAD COND	UNITY CHECK	LOAD COND
							Y-Y KSI	Z-Z KSI									
1-	2	FLG	0.999	C<.15	CMB1	30.00	-2.82	0.00	22.95	0.62	0.00	0.85	0.85	1.00	CMB2	0.00	
2-	3	FLG	0.851	C<.15	CMB1	0.00	-0.01	0.00	22.95	1.09	0.00	0.85	0.85	0.85	CMB2	0.00	
1001-	2	CBL	0.972	TN+BN	CMB1	0.00	20.99	0.00	0.00	0.00	0.00	0.85	0.85	0.01	CMB2	0.00	
1002-	2	CBL	0.972	TN+BN	CMB2	0.00	20.99	0.00	0.00	0.00	0.00	0.85	0.85	0.01	CMB1	0.00	
1003-	2	CBL	0.972	TN+BN	CMB1	0.00	20.99	0.00	0.00	0.00	0.00	0.85	0.85	0.01	CMB2	0.00	
1004-	2	CBL	0.972	TN+BN	CMB2	0.00	20.99	0.00	0.00	0.00	0.00	0.85	0.85	0.01	CMB1	0.00	

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen

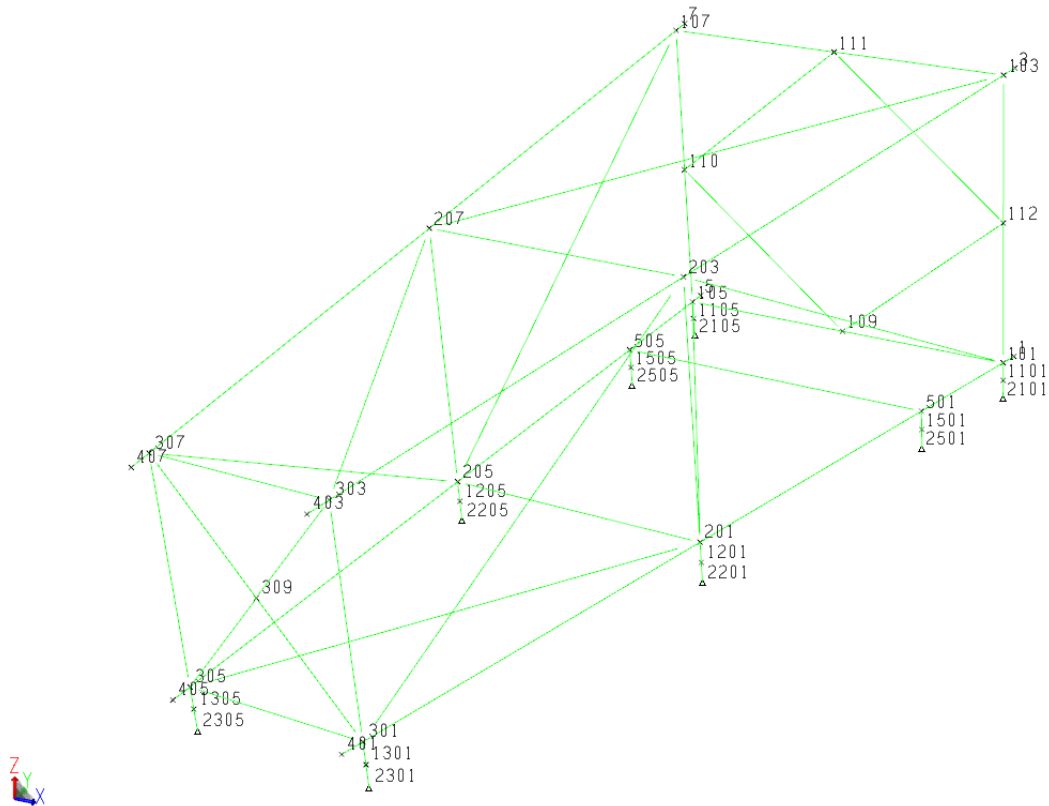
SACS-IV SYSTEM MEMBER INTERNAL LOADS SUMMARY REPORT

CASES		MAX. CRIT	LOAD	DIST	I N T E R N A L L O A D S							NEXT TWO HIGHEST		
MEMBER	GRP	UNITY COND	COND	FROM	AXIAL	SHEAR	SHEAR	TORSION	BENDING	BENDING	UNITY	LD	UNITY	
LD		CHECK	NO.	END		Y	Z		Y-Y	Z-Z	CHECK	CN	CHECK	
CN					FT	KIPS	KIPS	KIPS	IN-KIP	IN-KIP	IN-KIP			
1-	2 FLG	1.00 C<.15	CMB1	30.0	-129.21	14.275	0.0000	-0.33690E-18	-0.30943E-12	6000.0	1.0	CMB2	0.0	
2-	3 FLG	0.85 C<.15	CMB1	0.0	-0.66000	-25.000	0.0000	-0.24571E-34	0.0000	6000.0	0.9	CMB2	0.0	
1001-	2 CBL	0.97 TN+BN	CMB1	0.0	78.721	0.0000	0.0000	-0.59019E-18	0.23607E-17	-0.59019E-18	0.0	CMB2	0.0	
1002-	2 CBL	0.97 TN+BN	CMB2	0.0	78.721	0.0000	0.0000	-0.23665E-17	-0.29625E-17	0.65036E-17	0.0	CMB1	0.0	
1003-	2 CBL	0.97 TN+BN	CMB1	0.0	78.721	0.0000	0.0000	-0.23665E-17	-0.29625E-17	0.65036E-17	0.0	CMB2	0.0	
1004-	2 CBL	0.97 TN+BN	CMB2	0.0	78.721	0.0000	0.0000	0.17764E-17	-0.29625E-17	-0.41429E-17	0.0	CMB1	0.0	

4.2 SAMPLE PROBLEM 2 – JACKET TYPE STRUCTURE LOAD OUT

Sample Problem 2 is a simulation of a load out of a jacket type structure. The structure is supported by four pairs of supports located at the leg hard points. In addition to checking the structure when fully supported, the structure is to be checked when each of the pairs of supports is considered ineffective (i.e. located between the bulkhead and the barge or not touching because of change in barge deck elevation). For this sample, gap element data was specified directly in the model. A Gap input file was used to override the gap element type of each of the support pairs to 'no load' for a particular load case.

Joint Label: Name



The structure is modeled such that joints 2101, 2105, 2201, 2205, 2301, 2305, 2501 and 2505 are support points. Each of the load out cans are divided into two members. The bottom member attached to the support joint is designated as a compression only gap element.

The weight of unmodeled items such as walkways, lifting eyes and mudmats were accounted for in Load case 'MISC' of the SACS model while load case 'DEAD' contains the structure self-weight.

Load combinations 'LOS1', 'LOS2', 'LOS3' and 'LOS5' contain load cases Dead and MISC and are used to simulate the loss of support at joint pairs. Combination 'LOS1' will be used to represent loss of supports 2101 and 2105, while 'LOS2' will represent loss of 2201 and 2205, 'LOS3' loss of 2301 and 2305 and 'LOS5' loss of 2501 and 2505.

A portion of the SACS model file used for Sample Problem 2 along with a description of some of the input follows:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	GAP SAMPLE PROBLEM 2							
3	OPTIONS	EN	SDUC	2 1	DC	C	PT	PTPT
4	LCSEL	GP	ALLS	LOS1	LOS2	LOS3	LOS5	
5	GRUP	GAP	24.000	1.000	29.0111	.2036	.00	1
6	GRUP	LG1	41.339	1.378	29.0111	.2050	.00	1
7	GRUP	LG2	41.339	1.378	29.0111	.2050	.00	1
8	GRUP	LG2	40.551	0.984	29.0111	.2036	.00	1
9	GRUP	LG2	41.339	1.378	29.0111	.2050	.00	1
10	GRUP	LG3	41.339	1.378	29.0111	.2050	.00	1
11	GRUP	LG3	40.551	0.984	29.0111	.2036	.00	1
12	GRUP	LG3	41.339	1.378	29.0111	.2050	.00	1
13	GRUP	LG4	41.339	1.378	29.0111	.2050	.00	1
14	GRUP	T01	15.748	0.689	29.0111	.2036	.00	1
15	GRUP	T02	19.685	0.787	29.0111	.2036	.00	1
16	GRUP	T03	11.811	0.492	29.0111	.2036	.00	1
17	MEMBER							
18	MEMBER	1101101	GAP					
19	MEMBER	1105105	GAP					
20	MEMBER	1201201	GAP					
21	MEMBER	1205205	GAP					
22	MEMBER	1301301	GAP					
23	MEMBER	1305305	GAP					
24	MEMBER	1501501	GAP					
25	MEMBER	1505505	GAP					
26	MEMBER	21011101	GAP	C				
27	MEMBER	21051105	GAP	C				
28	MEMBER	22011201	GAP	C				
29	MEMBER	22051205	GAP	C				
30	MEMBER	23011301	GAP	C				
31	MEMBER	23051305	GAP	C				
32	MEMBER	25011501	GAP	C				
33	MEMBER	25051505	GAP	C				
34	***** ADDITIONAL MEMBER DATA *****							
35	JOINT							
36	***** ADDITIONAL JOINT DATA *****							
37	JOINT	2101	38.550164.042	-8.000				111000
38	JOINT	2105	-38.550164.042	-8.000				111000
39	JOINT	2201	26.657	68.898	-8.000			111000
40	JOINT	2205	-26.657	68.898	-8.000			111000
41	JOINT	2301	17.224	-6.562	-8.000			111000
42	JOINT	2305	-17.224	-6.562	-8.000			111000
43	JOINT	2501	34.920135.000	-8.000				111000
44	JOINT	2505	-34.920135.000	-8.000				111000
45	LOAD							
46	LOADCNDEAD							
47	***** ADDITIONAL DEAD LOADING *****							
48	LOADCNMISC							
49	LOAD	112		-4.0000			GLOB JOIN	MUDMAT
50	LOAD	109		-4.0000			GLOB JOIN	MUDMAT
51	LOAD	110		-4.0000			GLOB JOIN	MUDMAT
52	LOAD	111		-4.0000			GLOB JOIN	MUDMAT
53	LCOMB							
54	LCOMB	ALLS	MISC1.0000DEAD1.0000					
55	LCOMB	LOS1	MISC1.0000DEAD1.0000					
56	LCOMB	LOS2	MISC1.0000DEAD1.0000					
57	LCOMB	LOS3	MISC1.0000DEAD1.0000					
58	LCOMB	LOS5	MISC1.0000DEAD1.0000					
59	END							

-
- 3 Load case 'ALLS', 'LOS1', 'LOS2', 'LOS3' and 'LOS5' are selected for output for this gap analysis.
- 26 Members 2101-1101, 2105-1105, 2201-1105, 2205-1105, 2301-1301, 2305-1305, 2501-1501 and 2505-1505 are assigned to group GAP and are designated as compression only gap elements by 'C' in column 22.
- 37 Joints 2101, 2105, 2201, 2205, 2301, 2305, 2501 and 2505 are assigned as pinned supports ('111' in columns 55-57).
- 46 Dead load or the self-weight of the structure is represented in load case 'DEAD'.
- 48 Load case 'MISC' representing miscellaneous or unmodeled steel is defined.
- 54 The load combination 'ALLS' is used to represent the case when all supports are active.
- 55 The load combinations used to represent the loss of support are defined using the LCOMB lines. Load cases 'LOS1', 'LOS2', 'LOS3' and 'LOS4' contain load cases 'DEAD' and 'MISC'.

Below is the Gap input file used for this gap analysis. A detailed description of the input file follows:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPOPT	EN						
2	LCGAP	LOS1	INC	NL	MEM	21011101	21051105	
3	LCGAP	LOS2	INC	NL	MEM	22011201	22051205	
4	LCGAP	LOS3	INC	NL	MEM	23011301	23051305	
5	LCGAP	LOS5	INC	NL	MEM	25011501	25051505	
6	END							

The GAPOPT input designates the analysis options as follows:

- a. English units are designated in columns 21-22.
- 1 The first LCGAP line designates gap type overrides as follows:
 - a. Members 2101-1101 and 2105-1105 are changed to no load gap elements as designated by 'NL' in columns 16-17.
 - b. This override applies to load case 'LOS1'.
 - 2 The second LCGAP line designates gap type overrides as follows:
 - a. Members 2201-1201 and 2205-1205 are changed to no load gap elements as designated by 'NL' in columns 16-17.
 - b. This override applies to load case 'LOS2'.
 - 3 The third LCGAP line designates gap type overrides as follows:
 - a. Members 2301-1301 and 2305-1305 are changed to no load gap elements as designated by 'NL' in columns 16-17.
-

-
- b. This override applies to load case 'LOS3'.

1 The last LCGAP line designates gap type overrides as follows:

- a. Members 2501-1501 and 2505-1505 are changed to no load gap elements as designated by 'NL' in columns 16-17.
- b. This override applies to load case 'LOS5'.

The following is a portion of the output listing file created by the Gap analysis:

SACS CONNECT Edition (v11.0)
GAP SAMPLE PROBLEM 2

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:41:52 PRE PAGE 1

PRE VERSION 12.1.0.14

** PROGRAM OPTIONS **

UNITS

....ENGLISH

EXECUTION

....GAP ANALYSIS
....SHEAR DEFORMATION INCLUDED
....UNITY CHECK
WSD AISC 9TH EDITION WITH API-RP2A 21ST EDITION FOR TUBULARS
....DKT PLATES SELECTED
....NO SEGMENTS FOR PRISMATIC MEMBERS 2
....NO SEGMENTS/SECTION FOR NON-PRISMATIC MEMBERS 1

REPORTS SELECTED

....ELEMENT STRESS AT MAXIMUM UNITY CHECK...PRINT
....JOINT S.....PRINT

LOAD

....NO. BASIC LOAD COND. 2
....NO. COMB. LOAD COND. 5

SACS CONNECT Edition (v11.0)
GAP SAMPLE PROBLEM 2

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 15-AUG-2018 TIME 15:41:53 GAP PAGE 3

***** GAP ELEMENT REPORT *****

GAP ELEMENT	1ST JOINT	2ND JOINT	GRUP ID	LOAD CASE	ELEMENT TYPE	DEGREE OF FREEDOM	PRESET GAP IN	**** FRICTION ELEMENT **** COEFF. FRICT.	NORMAL J1	MEMBER J2	LOAD TYPE	1ST POINT FORCE KIPS	2ND POINT DEFL. IN	FORCE KIPS	DEFL. IN
1	2101	1101	GAP	G001	COMP	AXIAL	0.00								
2	2105	1105	GAP	G002	COMP	AXIAL	0.00								
3	2201	1201	GAP	G003	COMP	AXIAL	0.00								
4	2205	1205	GAP	G004	COMP	AXIAL	0.00								
5	2301	1301	GAP	G005	COMP	AXIAL	0.00								
6	2305	1305	GAP	G006	COMP	AXIAL	0.00								
7	2501	1501	GAP	G007	COMP	AXIAL	0.00								
8	2505	1505	GAP	G008	COMP	AXIAL	0.00								
LOAD CASE ALLS CONVERGED IN					0 ITERATIONS										

SACS CONNECT Edition (v11.0)			Bentley Systems		ID=Ym5ng2NpaIJxrIF8nIen		
GAP SAMPLE PROBLEM 2			DATE 15-AUG-2018			TIME 15:41:53	GAP PAGE 4
NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR		
1	2101-1101	COMP	-0.00197	-86.0	0.000000E+00		
2	2105-1105	COMP	-0.00180	-78.8	0.000000E+00		
3	2201-1201	COMP	-0.00255	-111.6	0.000000E+00		
4	2205-1205	COMP	-0.00292	-127.4	0.000000E+00		
5	2301-1301	COMP	-0.00128	-55.9	0.000000E+00		
6	2305-1305	COMP	-0.00113	-49.3	0.000000E+00		
7	2501-1501	COMP	-0.00072	-31.3	0.000000E+00		
8	2505-1505	COMP	-0.00073	-31.9	0.000000E+00		
LOAD CASE LOS1 CONVERGED IN 30 ITERATIONS							

SACS CONNECT Edition (v11.0)			Bentley Systems		ID=Ym5ng2NpaIJxrIF8nIen		
GAP SAMPLE PROBLEM 2			DATE 15-AUG-2018			TIME 15:41:53	GAP PAGE 5
NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR		
1	2101-1101	NULL	-1.00361	0.0	0.438277E+05		
2	2105-1105	NULL	-0.99521	0.0	0.434608E+05		
3	2201-1201	COMP	-0.00444	-193.7	0.000000E+00		
4	2205-1205	COMP	-0.00454	-198.1	0.000000E+00		
5	2301-1301	COMP	0.30220	0.0	-0.131972E+05		
6	2305-1305	COMP	0.32018	0.0	-0.139825E+05		
7	2501-1501	COMP	-0.00209	-91.1	0.000000E+00		
8	2505-1505	COMP	-0.00204	-89.2	0.000000E+00		
LOAD CASE LOS2 CONVERGED IN 30 ITERATIONS							

SACS CONNECT Edition (v11.0)			Bentley Systems		ID=Ym5ng2NpaIJxrIF8nIen		
GAP SAMPLE PROBLEM 2			DATE 15-AUG-2018			TIME 15:41:53	GAP PAGE 6
NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR		
1	2101-1101	COMP	-0.00300	-131.2	0.000000E+00		
2	2105-1105	COMP	-0.00291	-127.2	0.000000E+00		
3	2201-1201	NULL	-0.32097	0.0	0.140168E+05		
4	2205-1205	NULL	-0.37339	0.0	0.163060E+05		
5	2301-1301	COMP	-0.00267	-116.5	0.000000E+00		
6	2305-1305	COMP	-0.00274	-119.5	0.000000E+00		
7	2501-1501	COMP	-0.00084	-36.7	0.000000E+00		
8	2505-1505	COMP	-0.00094	-41.0	0.000000E+00		

LOAD CASE LOS3 CONVERGED IN 30 ITERATIONS

SACS CONNECT Edition (v11.0)
GAP SAMPLE PROBLEM 2

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	2101-1101	COMP	-0.00097	-42.2	0.000000E+00
2	2105-1105	COMP	-0.00086	-37.5	0.000000E+00
3	2201-1201	COMP	-0.00477	-208.1	0.000000E+00
4	2205-1205	COMP	-0.00501	-218.7	0.000000E+00
5	2301-1301	NULL	-0.50522	0.0	0.220631E+05
6	2305-1305	NULL	-0.49077	0.0	0.214319E+05
7	2501-1501	COMP	-0.00077	-33.5	0.000000E+00
8	2505-1505	COMP	-0.00073	-32.1	0.000000E+00
LOAD CASE LOS5 CONVERGED IN 29 ITERATIONS					

SACS CONNECT Edition (v11.0)
GAP SAMPLE PROBLEM 2

Bentley Systems

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SACS-IV SYSTEM FIXED JOINTS REACTION FORCES AND MOMENTS

JOINT NUMBER	LOAD CASE	***** KIPS *****			***** FT-KIPS *****		
		FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)
2101	ALLS	-2.916	-4.794	85.987	0.000	0.000	0.000
	LOS1	2.790	47.374	0.001	0.000	0.000	0.000
	LOS2	-3.515	-9.057	131.159	0.000	0.000	0.000
	LOS3	-1.893	4.062	42.171	0.000	0.000	0.000
	LOS5	-5.414	-28.376	109.367	0.000	0.000	0.000
2105	ALLS	5.712	-4.964	78.771	0.000	0.000	0.000
	LOS1	-0.516	42.173	0.001	0.000	0.000	0.000
	LOS2	5.264	-2.867	127.215	0.000	0.000	0.000
	LOS3	5.309	-1.116	37.463	0.000	0.000	0.000
	LOS5	9.022	-29.253	100.743	0.000	0.000	0.000
2201	ALLS	-10.577	-5.656	111.557	0.000	0.000	0.000
	LOS1	-15.179	-56.088	193.679	0.000	0.000	0.000
	LOS2	-10.635	-6.712	0.001	0.000	0.000	0.000
	LOS3	-11.158	-10.557	208.092	0.000	0.000	0.000
	LOS5	-8.420	19.233	118.220	0.000	0.000	0.000

2205	ALLS	4.640	-5.895	127.420	0.000	0.000	0.000
	LOS1	11.085	-52.664	198.122	0.000	0.000	0.000
	LOS2	4.811	-9.330	0.001	0.000	0.000	0.000
	LOS3	5.136	-8.503	218.694	0.000	0.000	0.000
	LOS5	1.298	18.708	136.631	0.000	0.000	0.000
2301	ALLS	-0.139	16.186	55.876	0.000	0.000	0.000
	LOS1	-2.839	-5.453	0.000	0.000	0.000	0.000
	LOS2	1.683	30.197	116.470	0.000	0.000	0.000
	LOS3	-1.665	2.650	0.001	0.000	0.000	0.000
	LOS5	0.481	20.814	56.618	0.000	0.000	0.000
2305	ALLS	3.215	15.166	49.285	0.000	0.000	0.000
	LOS1	4.050	-1.083	0.000	0.000	0.000	0.000
	LOS2	2.860	25.898	119.485	0.000	0.000	0.000
	LOS3	3.669	6.435	0.000	0.000	0.000	0.000
	LOS5	2.946	19.511	50.435	0.000	0.000	0.000
2501	ALLS	-7.129	-4.912	31.263	0.000	0.000	0.000
	LOS1	-4.687	14.750	91.062	0.000	0.000	0.000
	LOS2	-8.480	-15.674	36.664	0.000	0.000	0.000
	LOS3	-5.834	5.430	33.505	0.000	0.000	0.000
	LOS5	-7.763	-10.027	0.001	0.000	0.000	0.000
2505	ALLS	7.194	-5.131	31.856	0.000	0.000	0.000
	LOS1	5.296	10.990	89.151	0.000	0.000	0.000
	LOS2	8.012	-12.453	41.019	0.000	0.000	0.000
	LOS3	6.434	1.598	32.089	0.000	0.000	0.000
	LOS5	7.851	-10.609	0.001	0.000	0.000	0.000

SACS-IV SYSTEM MEMBER FORCES AND MOMENTS										
				***** KIPS *****				***** IN-KIPS *****		
MEMBER	MEMBER	GROUP	LOAD	FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)	
NUMBER	END	ID	CASE							
1101-	101	1101	GAP	ALLS	-85.00	-2.92	-4.79	0.00	-230.13	-139.97
				LOS1	0.98	2.79	47.37	0.00	2273.96	133.91
				LOS2	-130.18	-3.52	-9.06	0.00	-434.75	-168.72
				LOS3	-41.19	-1.89	4.06	0.00	194.99	-90.85
				LOS5	-108.38	-5.41	-28.38	0.00	-1362.06	-259.85
	101			ALLS	-84.02	-2.92	-4.79	0.00	-460.26	-279.95
				LOS1	1.97	2.79	47.37	0.00	4547.92	267.82
				LOS2	-129.19	-3.52	-9.06	0.00	-869.50	-337.44
				LOS3	-40.20	-1.89	4.06	0.00	389.99	-181.71
				LOS5	-107.40	-5.41	-28.38	0.00	-2724.12	-519.71

1105-	105	1105	GAP	ALLS	-77.79	5.71	-4.96	0.00	-238.28	274.16
				LOS1	0.98	-0.52	42.17	0.00	2024.31	-24.78
				LOS2	-126.23	5.26	-2.87	0.00	-137.62	252.69
				LOS3	-36.48	5.31	-1.12	0.00	-53.57	254.86
				LOS5	-99.76	9.02	-29.25	0.00	-1404.14	433.03
		105		ALLS	-76.80	5.71	-4.96	0.00	-476.56	548.32
				LOS1	1.97	-0.52	42.17	0.00	4048.62	-49.57
				LOS2	-125.25	5.26	-2.87	0.00	-275.24	505.38
				LOS3	-35.50	5.31	-1.12	0.00	-107.15	509.71
				LOS5	-98.78	9.02	-29.25	0.00	-2808.28	866.07
1201-	201	1201	GAP	ALLS	-110.57	-10.58	-5.66	0.00	-271.50	-507.70
				LOS1	-192.70	-15.18	-56.09	0.00	-2692.21	-728.60
				LOS2	0.98	-10.64	-6.71	0.00	-322.20	-510.50
				LOS3	-207.11	-11.16	-10.56	0.00	-506.73	-535.60
				LOS5	-117.24	-8.42	19.23	0.00	923.16	-404.18
		201		ALLS	-109.59	-10.58	-5.66	0.00	-543.00	-1015.41
				LOS1	-191.71	-15.18	-56.09	0.00	-5384.43	-1457.20
				LOS2	1.97	-10.64	-6.71	0.00	-644.40	-1021.00
				LOS3	-206.12	-11.16	-10.56	0.00	-1013.47	-1071.20
				LOS5	-116.25	-8.42	19.23	0.00	1846.32	-808.35
1205-	205	1205	GAP	ALLS	-126.44	4.64	-5.89	0.00	-282.95	222.73
				LOS1	-197.14	11.09	-52.66	0.00	-2527.86	532.08
				LOS2	0.98	4.81	-9.33	0.00	-447.86	230.93
				LOS3	-217.71	5.14	-8.50	0.00	-408.14	246.54
				LOS5	-135.65	1.30	18.71	0.00	897.97	62.30
		205		ALLS	-125.45	4.64	-5.89	0.00	-565.90	445.47
				LOS1	-196.15	11.09	-52.66	0.00	-5055.72	1064.16
				LOS2	1.97	4.81	-9.33	0.00	-895.72	461.86
				LOS3	-216.73	5.14	-8.50	0.00	-816.28	493.08
				LOS5	-134.66	1.30	18.71	0.00	1795.94	124.60
1301-	301	1301	GAP	ALLS	-54.89	-0.14	16.19	0.00	776.95	-6.67
				LOS1	0.98	-2.84	-5.45	0.00	-261.75	-136.27
				LOS2	-115.49	1.68	30.20	0.00	1449.44	80.79
				LOS3	0.98	-1.66	2.65	0.00	127.22	-79.90
				LOS5	-55.63	0.48	20.81	0.00	999.05	23.07
		301		ALLS	-53.91	-0.14	16.19	0.00	1553.90	-13.33
				LOS1	1.97	-2.84	-5.45	0.00	-523.50	-272.54
				LOS2	-114.50	1.68	30.20	0.00	2898.87	161.59
				LOS3	1.97	-1.66	2.65	0.00	254.43	-159.80
				LOS5	-54.65	0.48	20.81	0.00	1998.10	46.14
SACS-IV SYSTEM MEMBER FORCES AND MOMENTS										
***** KIPS *****										
***** IN-KIPS *****										
MEMBER	MEMBER	GROUP	LOAD	FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)	
NUMBER	END	ID	CASE							
2201-1201	2201	GAP	ALLS	-111.56	-10.58	-5.66	0.00	0.00	0.00	
			LOS1	-193.68	-15.18	-56.09	0.00	0.00	0.00	

			LOS2	0.00	-10.64	-6.71	0.00	0.00	0.00
			LOS3	-208.09	-11.16	-10.56	0.00	0.00	0.00
			LOS5	-118.22	-8.42	19.23	0.00	0.00	0.00
	1201		ALLS	-110.57	-10.58	-5.66	0.00	-271.50	-507.70
			LOS1	-192.70	-15.18	-56.09	0.00	-2692.21	-728.60
			LOS2	0.98	-10.64	-6.71	0.00	-322.20	-510.50
			LOS3	-207.11	-11.16	-10.56	0.00	-506.73	-535.60
			LOS5	-117.24	-8.42	19.23	0.00	923.16	-404.18
2205-1205	2205	GAP	ALLS	-127.42	4.64	-5.89	0.00	0.00	0.00
			LOS1	-198.12	11.09	-52.66	0.00	0.00	0.00
			LOS2	0.00	4.81	-9.33	0.00	0.00	0.00
			LOS3	-218.69	5.14	-8.50	0.00	0.00	0.00
			LOS5	-136.63	1.30	18.71	0.00	0.00	0.00
	1205		ALLS	-126.44	4.64	-5.89	0.00	-282.95	222.73
			LOS1	-197.14	11.09	-52.66	0.00	-2527.86	532.08
			LOS2	0.98	4.81	-9.33	0.00	-447.86	230.93
			LOS3	-217.71	5.14	-8.50	0.00	-408.14	246.54
			LOS5	-135.65	1.30	18.71	0.00	897.97	62.30
2301-1301	2301	GAP	ALLS	-55.88	-0.14	16.19	0.00	0.00	0.00
			LOS1	0.00	-2.84	-5.45	0.00	0.00	0.00
			LOS2	-116.47	1.68	30.20	0.00	0.00	0.00
			LOS3	0.00	-1.66	2.65	0.00	0.00	0.00
			LOS5	-56.62	0.48	20.81	0.00	0.00	0.00
	1301		ALLS	-54.89	-0.14	16.19	0.00	776.95	-6.67
			LOS1	0.98	-2.84	-5.45	0.00	-261.75	-136.27
			LOS2	-115.49	1.68	30.20	0.00	1449.44	80.79
			LOS3	0.98	-1.66	2.65	0.00	127.22	-79.90
			LOS5	-55.63	0.48	20.81	0.00	999.05	23.07
2305-1305	2305	GAP	ALLS	-49.28	3.22	15.17	0.00	0.00	0.00
			LOS1	0.00	4.05	-1.08	0.00	0.00	0.00
			LOS2	-119.49	2.86	25.90	0.00	0.00	0.00
			LOS3	0.00	3.67	6.43	0.00	0.00	0.00
			LOS5	-50.43	2.95	19.51	0.00	0.00	0.00
	1305		ALLS	-48.30	3.22	15.17	0.00	727.98	154.34
			LOS1	0.98	4.05	-1.08	0.00	-51.97	194.39
			LOS2	-118.50	2.86	25.90	0.00	1243.08	137.26
			LOS3	0.98	3.67	6.43	0.00	308.86	176.12
			LOS5	-49.45	2.95	19.51	0.00	936.55	141.40
2501-1501	2501	GAP	ALLS	-31.26	-7.13	-4.91	0.00	0.00	0.00
			LOS1	-91.06	-4.69	14.75	0.00	0.00	0.00
			LOS2	-36.66	-8.48	-15.67	0.00	0.00	0.00
			LOS3	-33.51	-5.83	5.43	0.00	0.00	0.00
			LOS5	0.00	-7.76	-10.03	0.00	0.00	0.00
	1501		ALLS	-30.28	-7.13	-4.91	0.00	-235.80	-342.21
			LOS1	-90.08	-4.69	14.75	0.00	708.01	-224.95
			LOS2	-35.68	-8.48	-15.67	0.00	-752.36	-407.02
			LOS3	-32.52	-5.83	5.43	0.00	260.66	-280.01

LOS5	0.98	-7.76	-10.03	0.00	-481.31	-372.64
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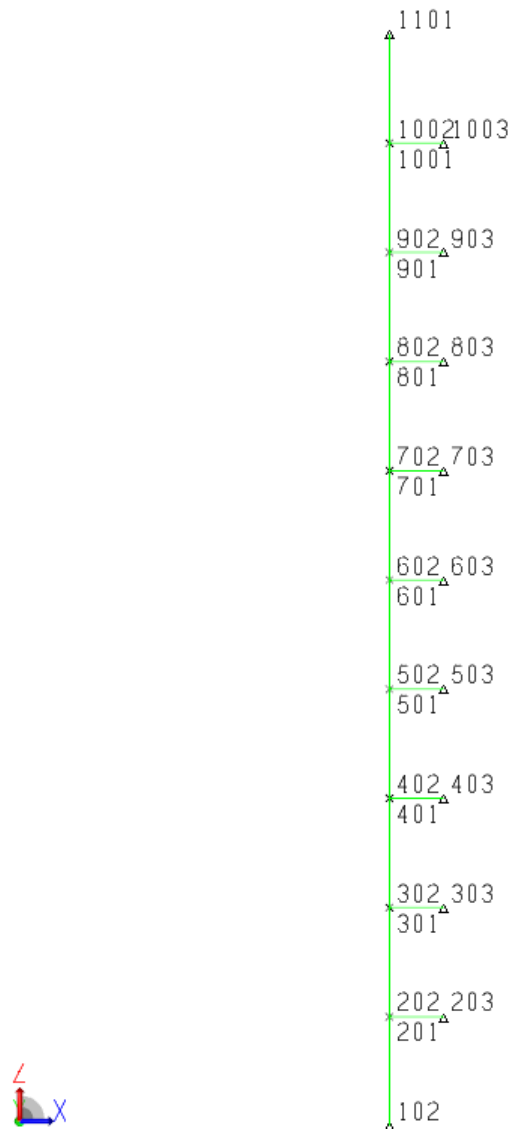
SACS CONNECT Edition (v11.0)				Bentley Systems				ID=Ym5ng2NpaIJxrIF8nIen									
GAP SAMPLE PROBLEM 2				DATE 15-AUG-2018				TIME 15:41:53				PST PAGE		6			
SACS-IV SYSTEM ELEMENT STRESS REPORT AT MAXIMUM UNITY CHECK																	
MEMBER		GRP	MAXIMUM UNITY CHECK	CRITICAL COND.	LOAD CASE NO.	DIST FROM END FT	***** AXIAL	APPLIED STRESSES *****			* CM VALUES *		* NEXT TWO HIGHEST CASES *				
								** BENDING **	*** SHEAR ***					UNITY CHECK	LOAD COND	UNITY CHECK	LOAD COND
							KSI	Y-Y KSI	Z-Z KSI	Y KSI	Z KSI	Y	Z				
1101-	101	GAP	0.424	TN+BN	LOS1	4.00	0.03	11.40	0.67	1.31	0.00	0.85	0.85	0.33	LOS5	0.17	LOS2
1105-	105	GAP	0.377	TN+BN	LOS1	4.00	0.03	10.15	-0.12	1.17	0.00	0.85	0.85	0.34	LOS5	0.13	LOS2
1201-	201	GAP	0.642	C<.15	LOS1	4.00	-2.65	-13.50	-3.65	1.61	0.00	0.85	0.85	0.27	LOS3	0.26	LOS5
1205-	205	GAP	0.607	C<.15	LOS1	4.00	-2.71	-12.67	2.67	1.49	0.00	0.85	0.85	0.25	LOS5	0.23	LOS3
1301-	301	GAP	0.344	C<.15	LOS2	4.00	-1.58	7.27	0.41	0.84	0.00	0.85	0.85	0.22	LOS5	0.18	ALLS
1305-	305	GAP	0.308	C<.15	LOS2	4.00	-1.63	6.23	0.69	0.72	0.00	0.85	0.85	0.21	LOS5	0.17	ALLS
1501-	501	GAP	0.196	C<.15	LOS1	4.00	-1.25	3.55	-1.13	0.43	0.00	0.85	0.85	0.18	LOS2	0.11	LOS5
1505-	505	GAP	0.166	C<.15	LOS1	4.00	-1.22	2.64	1.27	0.34	0.00	0.85	0.85	0.16	LOS2	0.12	LOS5
2101-	1101	GAP	0.212	TN+BN	LOS1	4.00	0.01	5.70	0.34	1.31	0.00	0.85	0.85	0.20	LOS5	0.13	LOS2
2105-	1105	GAP	0.201	C<.15	LOS5	4.00	-1.38	-3.52	1.09	0.85	0.00	0.85	0.85	0.19	LOS1	0.11	LOS2
2201-	1201	GAP	0.384	C<.15	LOS1	4.00	-2.67	-6.75	-1.83	1.61	0.00	0.85	0.85	0.20	LOS3	0.17	LOS5
2205-	1205	GAP	0.368	C<.15	LOS1	4.00	-2.73	-6.34	1.33	1.49	0.00	0.85	0.85	0.19	LOS3	0.17	LOS5
2301-	1301	GAP	0.210	C<.15	LOS2	4.00	-1.60	3.63	0.20	0.84	0.00	0.85	0.85	0.13	LOS5	0.11	ALLS
2305-	1305	GAP	0.193	C<.15	LOS2	4.00	-1.64	3.12	0.34	0.72	0.00	0.85	0.85	0.12	LOS5	0.10	ALLS
2501-	1501	GAP	0.127	C<.15	LOS1	4.00	-1.25	1.77	-0.56	0.43	0.00	0.85	0.85	0.10	LOS2	0.06	ALLS
2505-	1505	GAP	0.112	C<.15	LOS1	4.00	-1.22	1.32	0.64	0.34	0.00	0.85	0.85	0.09	LOS2	0.06	LOS5
1-	101	LG1	0.001	SHEAR	ALLS	4.03	0.00	-0.03	0.00	0.03	0.00	0.85	0.85	0.00	LOS1	0.00	LOS2
3-	103	LG1	0.001	SHEAR	ALLS	4.06	0.00	-0.03	0.00	0.03	0.00	0.85	0.85	0.00	LOS1	0.00	LOS2

5- 105 LG1	0.001	SHEAR	ALLS	4.03	0.00	-0.03	0.00	0.03	0.00	0.85	0.85	0.00	LOS1	0.00	LOS2
7- 107 LG1	0.001	SHEAR	ALLS	4.06	0.00	-0.03	0.00	0.03	0.00	0.85	0.85	0.00	LOS1	0.00	LOS2
101- 501 LG2	0.107	C<.15	LOS1	0.00	-0.13	3.85	-0.01	0.48	0.09	0.85	0.85	0.04	LOS5	0.03	LOS3
	0.244	C<.15	LOS1	24.38	-0.18	-6.35	0.00	0.86	0.13	0.85	0.85	0.10	LOS5	0.07	LOS2

4.3 SAMPLE PROBLEM 3 – RISER SIMULATION SUPPORTED BY GAP ELEMENTS

Sample Problem 3 is a simulation of a riser. The riser is pinned at each end and is supported by nine gap elements along its length. The riser supports are modeled by different types of gap elements: force-deflection, tension only, and compression only. All of the gap elements were defined in the model file to generate the dummy loads on the elements while the force-deflection elements and curves are defined in the gap input file.

Joint Label: Name



The structure is modeled such that joints 102, 203, 303, 403, 503, 603, 703, 803, 903, 1003, and 1103 are support points. Each riser support is connected to the riser via a wishbone element to limit the transfer of loading to axial loads.

A moment applied to the top of the riser is defined as load condition '1'. This load is factored in load combination '1000'. A series of loads distributed along the riser is defined as load condition '200'.

A portion of the SACS model file used for Sample Problem 3 along with a description of some of the input follows:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	GAP ELEMENT CHECK CASE							
3	OPTIONS EN UC 1 1 10 PTPTPT PT							
4	GRUP F-D	50.000	1.000	29.0011.2036.00	1	1.001.00		490.00
5	GRUP GAP	50.000	1.000	29.0011.2036.00	1	1.001.00		490.00
6	GRUP LG1	50.000	1.000	29.0011.2036.00		1.001.00		490.00
7	GRUP W.B	50.000	1.000	29.0011.2036.00		1.001.00		490.00
8	MEMBER							
9	MEMBER 601 603 F-D N							
10	MEMBER 701 703 F-D N							
11	MEMBER 201 203 GAP C							
12	MEMBER 301 303 GAP C							
13	MEMBER 401 403 GAP T							
14	MEMBER 501 503 GAP C							
15	MEMBER 801 803 GAP N							
16	MEMBER 901 903 GAP C							
17	MEMBER 10011003 GAP C							
18	MEMBER 102 202 LG1				0			
19	MEMBER 202 302 LG1				0			
20	MEMBER 302 402 LG1				0			
21	MEMBER 402 502 LG1				0			
22	MEMBER 502 602 LG1				0			
23	MEMBER 602 702 LG1				0			
24	MEMBER 702 802 LG1				0			
25	MEMBER 802 902 LG1				0			
26	MEMBER 902 1002 LG1				0			
27	MEMBER 10021101 LG1				0			
28	MEMBER1201 202 W.B		01111					
29	MEMBER OFFSETS			6.000				
30	MEMBER1301 302 W.B		01111					
31	MEMBER OFFSETS			6.000				
32	MEMBER1401 402 W.B		01111					
33	MEMBER OFFSETS			6.000				
34	MEMBER1501 502 W.B		01111					
35	MEMBER OFFSETS			6.000				
36	MEMBER1601 602 W.B		01111					
37	MEMBER OFFSETS			6.000				
38	MEMBER1701 702 W.B		01111					
39	MEMBER OFFSETS			6.000				
40	MEMBER1801 802 W.B		01111					
41	MEMBER OFFSETS			6.000				
42	MEMBER1901 902 W.B		01111					
43	MEMBER OFFSETS			6.000				
44	MEMBER110011002 W.B		01111					
45	MEMBER OFFSETS			6.000				
46	JOINT							
47	JOINT 102	0.	0.	0.		111001		
48	JOINT 201	0.	0.	10.				
49	JOINT 202	0.	0.	10.				
50	JOINT 203	5.	0.	10.		111111		
51	JOINT 301	0.	0.	20.				
52	JOINT 302	0.	0.	20.				
53	JOINT 303	5.	0.	20.		111111		

54	JOINT 401	0.	0.	30.				
55	JOINT 402	0.	0.	30.				
56	JOINT 403	5.	0.	30.	111111			
57	JOINT 501	0.	0.	40.				
58	JOINT 502	0.	0.	40.				
59	JOINT 503	5.	0.	40.	111111			
60	JOINT 601	0.	0.	50.				
61	JOINT 602	0.	0.	50.				
62	JOINT 603	5.	0.	50.	111111			
63	JOINT 701	0.	0.	60.				
64	JOINT 702	0.	0.	60.				
65	JOINT 703	5.	0.	60.	111111			
66	JOINT 801	0.	0.	70.				
67	JOINT 802	0.	0.	70.				
68	JOINT 803	5.	0.	70.	111111			
69	JOINT 901	0.	0.	80.				
70	JOINT 902	0.	0.	80.				
71	JOINT 903	5.	0.	80.	111111			
72	JOINT 1001	0.	0.	90.				
73	JOINT 1002	0.	0.	90.				
74	JOINT 1003	5.	0.	90.	111111			
75	JOINT 1101	0.	0.	100.	111000			
76	LOAD							
77	LOADCN 1							
78	LOAD 1101			20000.0	GLOB JOIN			
79	LOADCN 200							
80	LOAD X 102 202	20.0000	20.0000		GLOB UNIF	MEMBER		
81	LOAD X 202 302	20.0000	20.0000		GLOB UNIF	MEMBER		
82	LOAD X 302 402	20.0000	20.0000		GLOB UNIF	MEMBER		
83	LOAD X 402 502	20.0000	20.0000		GLOB UNIF	MEMBER		
84	LOAD X 502 602	20.0000	20.0000		GLOB UNIF	MEMBER		
85	LOAD X 602 702	20.0000	20.0000		GLOB UNIF	MEMBER		
86	LOAD X 702 802	20.0000	20.0000		GLOB UNIF	MEMBER		
87	LOAD X 802 902	20.0000	20.0000		GLOB UNIF	MEMBER		
88	LOAD X 902 1002	20.0000	20.0000		GLOB UNIF	MEMBER		
89	LOAD X 10021101	20.0000	20.0000		GLOB UNIF	MEMBER		
90	LCOMB							
91	LCOMB 1000 1	1.00+3						
92	END							

9-10 Members 601-603 and 701-703 are assigned to group F-D are defined as 'No Load' elements in column 22 so that the gap dummy loads are generated automatically by SACS.

11-17 Members 201-203, 301-303, 401-403, 501-503, 801-803, 901-903, and 1001-1003 are assigned to group GAP and are designated as gap elements in column 22.

28, 30, 32, 34, 36, 38, 40, 42, 44 Members 1201-202, 1301-302, 1401-402, 1501-502, 1601-602, 1701-702, 1801-802, 1901-902, and 1100-1102 are defined as assigned to group W.B and are defined as wishbone elements with Joint B end releases '01111 '.

47, 75 Joints 102 and 1101 at each end of the riser are pinned with fixity '111001' and '111000' respectively. Note that the fixity for the rotation about the Z axis ensures that the structure will be stable.

77 Load case '1' represents a moment applied at the top of the riser.

79 Load case '2' represents a distributed load along the length of the riser.

91 Load combination '1000' is load condition '1' factored by 1000.

Below is the Gap input file used for this gap analysis. A detailed description of the input file follows:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	GAPOPT	2	1	0	EN			
2	GAPELM	601	603		FD			
3	F-DEL	-100.	-10.0	-100.	-.02	0.0	-.01	0.0
4	F-DEL	+100.	0.02	+100.	10.000			0.01
5	GAPELM	701	703		RP			
6	END							

The GAPOPT input designates the analysis options as follows:

2 Member 601-603 is redefined as a Force-Deflection element, 'FD', in columns 24-25.

3, 4 The force deflection curve for element 601-603 is defined with the following points: (-100 kip, -10.0 in) (-100 kip, -0.02 in) (0.0 kip, -0.01 in) (0.0 kip, 0.01 in).

5 Member 701-703 is redefined as a Force-Deflection element with the previous element's properties using the Repeat option, 'RP' in columns 24-25.

The following is a portion of the output listing file created by the Gap analysis:

** PROGRAM OPTIONS **

UNITS

....ENGLISH

EXECUTION

....GAP ANALYSIS
....UNITY CHECK
WSD AISC 9TH EDITION WITH API-RP2A 21ST EDITION FOR TUBULARS
....DKT PLATES SELECTED
....NO SEGMENTS FOR PRISMATIC MEMBERS 1
....NO SEGMENTS/SECTION FOR NON-PRISMATIC MEMBERS 1

REPORTS SELECTED

....INTERPRETIVE INPUT.....PRINT
....INPUT ECHO.....PRINT
....JOINT DEFLECTIONS.....PRINT
....ELEMENT DETAIL.....PRINT
....SPECIAL ELEMENT.....PRINT

LOAD

....NO. BASIC LOAD COND. 2
....NO. COMB. LOAD COND. 1

***** GAP ELEMENT REPORT *****

GAP ELEMENT	1ST JOINT	2ND JOINT	GRP ID	LOAD CASE	ELEMENT TYPE	DEGREE OF FREEDOM	PRESET GAP IN	**** FRICTION ELEMENT ****				1ST POINT		2ND POINT	
								COEFF. FRICT.	NORMAL J1	MEMBER J2	LOAD TYPE	FORCE KIPS	DEFL. IN	FORCE KIPS	DEFL. IN
1	601	603	F-D	G001	F-D	AXIAL	0.00					-100.000	-10.000	-100.000	-
0.020												0.000	-0.010	0.000	
0.010												100.000	0.020	100.000	
10.000															

2	701	703	F-D	G002	F-D	AXIAL	0.00	-100.000	-10.000	-100.000	-
0.020								0.000	-0.010	0.000	
0.010								100.000	0.020	100.000	
10.000											
3	201	203	GAP	G003	COMP	AXIAL	0.00				
4	301	303	GAP	G004	COMP	AXIAL	0.00				
5	401	403	GAP	G005	TENS	AXIAL	0.00				
6	501	503	GAP	G006	COMP	AXIAL	0.00				
7	801	803	GAP	G007	NULL	AXIAL	0.00				
8	901	903	GAP	G008	COMP	AXIAL	0.00				
9	1001	1003	GAP	G009	COMP	AXIAL	0.00				
LOAD CASE 1 CONVERGED IN 30 ITERATIONS											

SACS CONNECT Edition (v11.0)
GAP ELEMENT CHECK CASE

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	601- 603	F-D	-0.00461	0.0	0.343020E+03
2	701- 703	F-D	0.01850	85.0	-0.129157E+04
3	201- 203	COMP	0.00000	0.0	-0.361796E+00
4	301- 303	COMP	-0.00002	-1.3	0.205846E-02
5	401- 403	TENS	0.00016	11.8	-0.117046E-02
6	501- 503	COMP	-0.00055	-41.0	0.206786E-02
7	801- 803	NULL	0.09007	0.0	-0.670142E+04
8	901- 903	COMP	0.15768	0.0	-0.117317E+05
9	1001-1003	COMP	0.15110	0.0	-0.112423E+05
LOAD CASE 200 CONVERGED IN 30 ITERATIONS					

SACS CONNECT Edition (v11.0)
GAP ELEMENT CHECK CASE

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	601- 603	F-D	-0.10712	-100.0	0.787043E+04
2	701- 703	F-D	-0.17567	-100.0	0.129708E+05
3	201- 203	COMP	-0.00278	-207.2	0.000000E+00
4	301- 303	COMP	-0.00333	-248.1	0.000000E+00
5	401- 403	TENS	-0.00437	0.0	0.324999E+03
6	501- 503	COMP	-0.00714	-530.9	0.000000E+00
7	801- 803	NULL	-0.12304	0.0	0.915466E+04
8	901- 903	COMP	-0.00809	-602.1	0.000000E+00
9	1001-1003	COMP	-0.00028	-20.9	0.000000E+00

LOAD CASE 1000 CONVERGED IN 34 ITERATIONS

SACS CONNECT Edition (v11.0)
GAP ELEMENT CHECK CASE

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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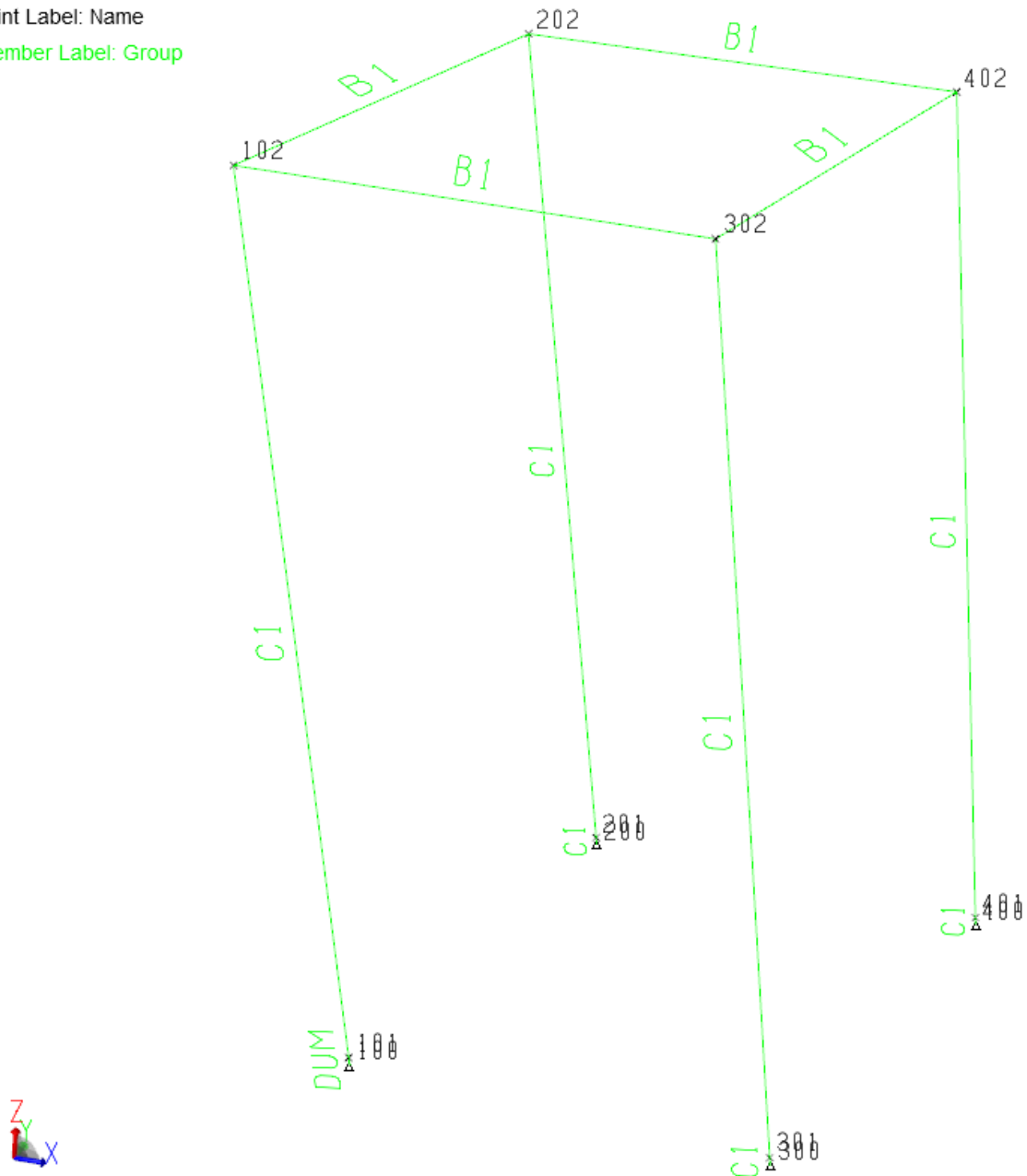
NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	601- 603	F-D	193.89450	100.0	-0.144263E+08
2	701- 703	F-D	320.21215	100.0	-0.238248E+08
3	201- 203	COMP	2.34291	0.0	-0.174320E+06
4	301- 303	COMP	-1.15856	-86200.7	0.798296E+00
5	401- 403	TENS	1.55356	115590.2	-0.805806E+00
6	501- 503	COMP	73.10300	0.0	-0.543911E+07
7	801- 803	NULL	408.07679	0.0	-0.303623E+08
8	901- 903	COMP	413.40162	0.0	-0.307585E+08
9	1001-1003	COMP	292.07877	0.0	-0.217316E+08

4.4 SAMPLE PROBLEM 4 – EQUIPMENT PLATFORM WITH FRICTION SUPPORT

Sample Problem 4 is a simulation of an equipment platform. One of the columns is supported by a friction element near the support.

Joint Label: Name

Member Label: Group



The structure is modeled with fixed connections at 100, 200, 300, and 400. Self-weight of the structure is applied as load condition 'DEAD'. Load condition '1' is a series of loads applied to the top of each column. Load condition '2' is a series of distributed loads along each column in the global X direction. Load condition '3' is a series of distributed loads along each column in the global Y direction. Load combinations '100', '200', and '300' are combinations of the 'Dead' load condition with load conditions '1', '2', and '3' respectively. Load combination '400' is a combination of the 'Dead' load condition with load conditions '2' and '3'.

A portion of the SACS model file used for Sample Problem 4 along with a description of some of the input follows:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	OPTIONS	EN	UC	1 1	DC	C		
2	LCSEL	100	200	300	400			
3	GRUP							
4	GRUP B1	W10X22		29.0011.2036.00	1	1.001.00		490.00
5	GRUP C1		8.000 0.125	29.0011.2036.00	1	1.001.00	0.500	490.00
6	GRUP DUM		8.000 0.125F	29.0011.2036.00	1	1.001.00	0.500	490.00
7	MEMBER							
8	MEMBER 102	202	B1					
9	MEMBER 102	302	B1					
10	MEMBER 202	402	B1					
11	MEMBER 302	402	B1					
12	MEMBER 101	102	C1					
13	MEMBER 200	201	C1					
14	MEMBER 201	202	C1					
15	MEMBER 300	301	C1					
16	MEMBER 301	302	C1					
17	MEMBER 400	401	C1					
18	MEMBER 401	402	C1					
19	MEMBER 100	101	DUM					
20	JOINT							
21	JOINT 100	0.	0.	0.		FIXED		
22	JOINT 101	0.	0.	0.	1.200			
23	JOINT 102	0.	0.	10.				
24	JOINT 200	0.	5.	0.		FIXED		
25	JOINT 201	0.	5.	0.	1.200			
26	JOINT 202	0.	5.	10.				
27	JOINT 300	5.	0.	0.		FIXED		
28	JOINT 301	5.	0.	0.	1.200			
29	JOINT 302	5.	0.	10.				
30	JOINT 400	5.	5.	0.		FIXED		
31	JOINT 401	5.	5.	0.	1.200			
32	JOINT 402	5.	5.	10.				
33	LOAD							
34	LOADCN 1							
35	LOAD 102	-10.000-10.000				GLOB JOIN	LATERAL	
36	LOAD 202	-10.000-10.000				GLOB JOIN	LATERAL	
37	LOAD 302	-10.000-10.000				GLOB JOIN	LATERAL	
38	LOAD 402	-10.000-10.000				GLOB JOIN	LATERAL	
39	LOADCN 2							
40	LOAD X 101 102	-1.0000		-1.0000		GLOB UNIF	LAT_X	
41	LOAD X 201 202	-1.0000		-1.0000		GLOB UNIF	LAT_X	
42	LOAD X 401 402	-1.0000		-1.0000		GLOB UNIF	LAT_X	
43	LOAD X 301 302	-1.0000		-1.0000		GLOB UNIF	LAT_X	
44	LOADCN 3							
45	LOAD Y 101 102	-0.5000		-0.5000		GLOB UNIF	LAT_Y	
46	LOAD Y 201 202	-0.5000		-0.5000		GLOB UNIF	LAT_Y	
47	LOAD Y 401 402	-0.5000		-0.5000		GLOB UNIF	LAT_Y	
48	LOAD Y 301 302	-0.5000		-0.5000		GLOB UNIF	LAT_Y	
49	LOADCNDEAD							
50	LOAD Z 102 302	-0.0221		-0.0221		GLOB UNIF	DEAD	
51	LOAD Z 202 402	-0.0221		-0.0221		GLOB UNIF	DEAD	
52	LOAD Z 102 202	-0.0221		-0.0221		GLOB UNIF	DEAD	

53	LOAD Z 302 402	-0.0221	-0.0221		GLOB UNIF	DEAD
54	LOAD Z 100 101	-0.0105	-0.0105		GLOB UNIF	DEAD
55	LOAD Z 101 102	-0.0105	-0.0105		GLOB UNIF	DEAD
56	LOAD Z 200 201	-0.0105	-0.0105		GLOB UNIF	DEAD
57	LOAD Z 201 202	-0.0105	-0.0105		GLOB UNIF	DEAD
58	LOAD Z 400 401	-0.0105	-0.0105		GLOB UNIF	DEAD
59	LOAD Z 401 402	-0.0105	-0.0105		GLOB UNIF	DEAD
60	LOAD Z 300 301	-0.0105	-0.0105		GLOB UNIF	DEAD
61	LOAD Z 301 302	-0.0105	-0.0105		GLOB UNIF	DEAD
62	LCOMB					
63	LCOMB 100 1	1.0000	DEAD1.0000			
64	LCOMB 200 2	1.0000	DEAD1.0000			
65	LCOMB 300 3	1.0000	DEAD1.0000			
66	LCOMB 400 2	1.0000	3 1.0000	DEAD1.0000		
67	END					

6 Group 'DUM' is defined as a friction element with the 'F' option in column 30.

100 Member 100-101 is defined as a 'DUM' element in columns 17-19.

21, 24, 27, 30 Joints 100, 200, 300, and 400 are defined as 'FIXED' in columns 55-60.

34, 39, 44, 49 Load case '1' represents lateral joint loads applied at the top of each column. Load case '2' represents a lateral distributed load in the global X direction on each column. Load case '3' represents a lateral distributed load in the global Y direction on each column. Load case 'DEAD' represents the dead load on the structure.

63, 64, 65, 66 Load combination '100', '200', and '300' are the combinations of the 'DEAD' load case and load cases '1', '2', and '3' respectively. Load combination '400' is the combination of the 'DEAD' load case and load cases '2' and '3'.

Below is the Gap input file used for this gap analysis. A detailed description of the input file follows:

	1	2	3	4	5	6	7	8
1	GAPOPT	0	EN 600	.5	.5			
2	END							

The GAPOPT input designates the analysis options as follows:

1 The friction coefficients for the Local Y and Local Z directions are defined as '0.5' and '0.5' in columns 35-39 and 40-44 respectively.

Note: The gap input file is not required for this analysis. As the input on the GAPOPT line can be entered in the Gap Analysis Options in the Gap Runfile. The gap input file is here for illustrative purposes.

The following is a portion of the output listing file created by the Gap analysis:

```

SACS CONNECT Edition (v11.0)          Bentley Systems          ID=Ym5ng2NpaIJxrIF8nIen
                                         DATE 16-AUG-2018  TIME 16:49:53  PRE PAGE    1
                                         PRE VERSION 12.1.0.14

** PROGRAM OPTIONS **

UNITS
....ENGLISH

EXECUTION
....GAP ANALYSIS
....UNITY CHECK
      WSD AISC 9TH EDITION WITH API-RP2A 21ST EDITION FOR TUBULARS
....DKT PLATES SELECTED
....NO SEGMENTS FOR PRISMATIC MEMBERS      1
....NO SEGMENTS/SECTION FOR NON-PRISMATIC MEMBERS      1

REPORTS SELECTED

LOAD
....NO. BASIC LOAD COND.      4
....NO. COMB. LOAD COND.      4

```

```

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***** GAP ELEMENT REPORT *****

      PRESET  **** FRICTION ELEMENT ****      1ST POINT      2ND POINT
      GAP    1ST  2ND  GRUP  LOAD  ELEMENT  DEGREE OF  GAP    COEFF.  NORMAL MEMBER LOAD  FORCE  DEFL.  FORCE  DEFL.
ELEMENT JOINT JOINT ID   CASE  TYPE      FREEDOM  IN    FRICT.   J1   J2   TYPE  KIPS   IN   KIPS   IN
      1      100  101  DUM   G001  COMP     AXIAL     0.00
      2      100  101  DUM   G002  FRIC     SHEAR-Y   0.00   0.500   100  101  COMP
      3      100  101  DUM   G003  FRIC     SHEAR-Z   0.00   0.500   100  101  COMP
      LOAD CASE 100 CONVERGED IN      0 ITERATIONS

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      NO.    GAP ELEMENT      TYPE      DEFLECTION      FORCE      FACTOR
              IN              KIPS

```

1	100- 101	COMP	-0.00052	-39.2	0.000000E+00
2	100- 101	FRIC	0.00001	10.0	0.000000E+00
3	100- 101	FRIC	0.00001	10.0	0.000000E+00
LOAD CASE 200 CONVERGED IN 30 ITERATIONS					

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	100- 101	COMP	-0.00001	-0.6	0.000000E+00
2	100- 101	FRIC	-29.47132	-0.2	-0.355800E+08
3	100- 101	FRIC	3.78044	0.2	0.456403E+07
LOAD CASE 300 CONVERGED IN 28 ITERATIONS					

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
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NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	100- 101	COMP	-0.00001	-0.4	0.000000E+00
2	100- 101	FRIC	2.01026	0.2	0.242694E+07
3	100- 101	FRIC	-14.85860	-0.2	-0.179384E+08
LOAD CASE 400 CONVERGED IN 29 ITERATIONS					

SACS CONNECT Edition (v11.0)

Bentley Systems

ID=Ym5ng2NpaIJxrIF8nIen
DATE 16-AUG-2018 TIME 16:49:54 GAP PAGE 7

NO.	GAP ELEMENT	TYPE	DEFLECTION IN	FORCE KIPS	FACTOR
1	100- 101	COMP	-0.00001	-0.6	0.000000E+00
2	100- 101	FRIC	-27.75943	-0.2	-0.335133E+08
3	100- 101	FRIC	-11.85554	-0.2	-0.143129E+08

5 INPUT LINES

END OF INPUT

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	THIS LINE IS THE LAST LINE IN THE 'GAP' INPUT FILE.
GENERAL	THE 'END' LINE TERMINATES THE DATA READ BY THE 'GAP' PROGRAM. THIS LINE IS OPTIONAL.

LINE LABEL	REMAINDER OF THIS INPUT LINE LEFT BLANK
END	
1-- 3	4-----80

FORCE VERSUS DEFLECTION CURVE DEFINITION

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS REQUIRED FOR EACH NONLINEAR GAP ELEMENT THAT HAS A 'FD' DESIGNATION. THIS DATA FOLLOWS DIRECTLY AFTER THE 'GAPELM' LINE DEFINING THE GAP ELEMENT TO WHICH THIS FORCE VERSUS DEFLECTION RELATIONSHIP APPLIES.
(9-26)	ENTER THE FIRST FORCE-DEFLECTION POINT TO DEFINE THIS CURVE.
(27-80)	ENTER THE SECOND, THIRD AND FOURTH POINTS AS REQUIRED. THESE POINTS MUST BE ENTERED IN ORDER OF INCREASING DEFLECTIONS. FOR DEFLECTIONS OUTSIDE THE RANGE OF THE TABLE VALUES, THE PROGRAM USES THE FIRST OR LAST FORCE VALUE IN THE TABLE AS APPROPRIATE. REPEAT THIS LINE AS OFTEN AS NECESSARY TO DEFINE THE FORCE VERSUS DEFLECTION CURVE IN AS MUCH DETAIL AS DESIRED. DO NOT USE BLANKS FOR ZEROS.

LINE LABEL	FIRST POINT		SECOND POINT		THIRD POINT		FOURTH POINT	
	FORCE	DEFL.	FORCE	DEFL.	FORCE	DEFL.	FORCE	DEFL.
F-DEL								
1-- 5	9<--17	18<--26	27<--35	36<--44	45<--53	54<--62	63<--71	72<--80
DEFAULT								
ENGLISH	KIP	IN	KIP	IN	KIP	IN	KIP	IN
METRIC (KN)	KN	CM	KN	CM	KN	CM	KN	CM
METRIC (KG)	KG	CM	KG	CM	KG	CM	KG	CM

'GAP' OPTIONS INPUT

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS REQUIRED IN ANY NONLINEAR GAP ELEMENT ANALYSIS. IT SPECIFIES OVERALL ANALYSIS PARAMETERS.
(7-10)	ENTER THE NUMBER OF REAL LOAD CASES IN THE SACS IV DATA. THE REAL LOAD CASES ARE THOSE THAT REPRESENT ACTUAL LOADING ON THE STRUCTURE AS OPPOSED TO THE DUMMY LOAD CASES THAT ARE USED TO SIMULATE THE GAP ELEMENTS.
(11-14)	ENTER THE NUMBER OF OUTPUT LOAD CASES. THESE ARE THE LOAD CASES THAT REPRESENT THE ACTUAL LOADING CONDITIONS TO BE ANALYZED AND IN GENERAL THESE WILL BE COMBINATIONS OF THE REAL LOAD CASES IN THE SACS IV DATA.
(15-18)	THIS OPTION IS PRIMARILY USED FOR DEBUGGING ANY PROBLEMS THAT MAY EXIST. IT WILL PRODUCE A GREAT DEAL OF OUTPUT THAT WILL NOT BE USEFUL FOR THE NORMAL CASE.
(19-20)	ENTER 'TW' IF THE AUTOMATIC 'TOW' ANALYSIS IS BEING USED. THIS OPTION REQUIRES THAT THE FIRST ACTUAL LOAD CASE IS A 'DEAD' LOAD CASE. THE FOLLOWING LOAD CASES ARE MOTION LOAD CASES WITHOUT GRAVITY. IF GRAVITY IS INCLUDED IN THE MOTION LOAD CASES, THEN ENTER 'TG' FOR THIS OPTION. THE TIE DOWNS ARE SPECIFIED IN THE SACS IV DATA AS NULL ELEMENTS.

COLUMNS	COMMENTARY
(21-22)	ENTER THE SYSTEM OF UNITS USED. THIS SYSTEM WILL NORMALLY BE THE SAME AS USED IN THE SACS IV ANALYSIS, BUT IS NOT REQUIRED TO BE SO. SELECT FROM ONE OF THE FOLLOWING: 'EN' - ENGLISH UNITS. 'MN' - METRIC UNITS WITH KILONEWTONS AS THE FORCE UNITS. 'ME' - METRIC UNITS WITH KILOGRAMS AS THE FORCE UNITS.
(23-26)	ENTER THE NUMBER OF ITERATIONS ALLOWED IN THE NONLINEAR SOLUTION.
(27-34)	ENTER THE CONVERGENCE TOLERANCE REQUIRED. THIS TOLERANCE SHOULD BE SMALL SINCE THE GAP ELEMENTS TYPICALLY ARE SHORT AND RELATIVELY STIFF.
(35-44)	ENTER THE FRICTION COEFFICIENTS FOR THE FRICTION ELEMENTS CREATED IN THE SACS IV INPUT FILE. THIS DATA CAN BE OVERRIDDEN FOR INDIVIDUAL ELEMENTS ON THE 'GAPELM' LINE.
(45-47)	ENTER 'PFD' OR 'PFG' TO PLOT FORCE VERSUS DEFLECTION FOR GAP ELEMENTS. 'PFD' WILL PLOT WITHOUT GRID LINES; 'PFG' WILL PLOT WITH GRID LINES. WITH BLANK INPUT NO PLOTS WILL BE PRODUCED.
(77-78)	LEAVE BLANK FOR THE STANDARD SOLVER (DEFAULT). ENTER 'A1' FOR ADVANCED MATRIX INVERSION,OR ENTER 'A2' FOR ADVANCED LINEAR SOLVER. ENTER 'AD' FOR THE DIRECT SOLVER. THE DIRECT SOLVER IS ONLY APPLICABLE TO MODELS WITH TENSION-ONLY, COMPRESSION-ONLY, AND/OR NO-LOAD ELEMENTS WITH ZERO INITIAL GAP VALUES. THE PROGRAM AUTOMATICALLY SWITCHES TO 'A2' IF IT CANNOT USE DIRECT SOLVER FOR ANY REASON.
(79-80)	ENTER 'NE' TO CONTINUE THE GAP ANALYSIS ANALYSIS IN THE CASE NON-CONVERGENCE WITH A WARNING MESSAGE INDICATING THE NON-CONVERGENCE. LEAVE BLANK FOR DEFAULT WORKFLOW (STOP WITH AN ERROR MESSAGE INDICATING THE NON-CONVERGENCE) (NOTE) THE RESULTS MUST BE DOUBLE-CHECKED IF THE NON-CONVERGENCE CONTINUE OPTION IS USED.

LINE LABEL	NUMBER OF REAL LOAD CASES	NUMBER OF OUTPUT LOAD CASES	DIAGNOSTIC OUTPUT PRINT OPTION	AUTOMATIC TOW ANALYSIS OPTION	UNITS OPTION	MAXIMUM NUMBER OF ITERATIONS	CONVERGENCE TOLERANCE	FRICTION COEFFICIENTS		FORCE DEFLECTION PLOT	LEAVE BLANK	ADAVNCED SOLVER	CONTINUE OPTION FOR NON-CONVERGENCE?
								LOCAL Y	LOCAL Z				
GAPOPT													
1-- 6	7-->10	11-->14	15-->18	19--20	21--22	23-->26	27<--34	35<--39	40<--44	45--47	48--76	77-78	79-80
DEFAULT					'EN'	1000	0.00001	0.5	0.5			STANDARD SOLVER	STOP IN THE CASE OF NON-CONVERGENCE

GROUP GAP TYPE OVERRIDE

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS OPTIONAL IN ANY NONLINEAR GAP ELEMENT ANALYSIS. IT ALLOWS THE USER TO CHANGE THE GAP ELEMENT TYPE OF A GROUP OF GAP ELEMENTS DEFINED IN THE MODEL FOR PARTICULAR LOAD CASES. THIS DATA CAN BE REPEATED AS NECESSARY TO SELECT ALL THE DESIRED GAP ELEMENTS.
(7-10)	ENTER THE LOAD CASE TO WHICH THESE OVERRIDES APPLY.
(12-14)	SELECT EITHER INCLUDE ('INC') TO SPECIFY THOSE GAP ELEMENT GROUPS THAT ARE TO BE MODIFIED FOR THIS LOAD CASE OR EXCLUDE ('EXC') TO SPECIFY THAT ALL GAP GROUPS ELEMENTS EXCEPT THOSE SPECIFIED ARE TO BE MODIFIED. NOTE THAT THE EXCLUDES AND INCLUDES CANNOT BE MIXED FOR ANY ONE LOAD CASE.
(16-17)	ENTER THE GAP ELEMENT TYPE FROM THE FOLLOWING CHOICES: (Default: 'CO')
(19-21)	OVERRIDE TYPE 'ST' - STANDARD SACS ELEMENT. 'CO' - COMPRESSION ONLY. 'TO' - TENSION ONLY. 'NL' - NO LOAD. 'FD' - FORCE VERSUS DEFLECTION ELEMENT (SEE 'F-DEL' DATA). 'RP' - REPEAT PREVIOUS GAP ELEMENT FORCE-DEFLECTION.
(23-25)	ENTER THE FIRST GROUP NAME.
(27-77)	ENTER THE REMAINING GROUP NAMES.

[illegible]

MEMBER GAP TYPE OVERRIDE

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS OPTIONAL IN ANY NONLINEAR GAP ELEMENT ANALYSIS. IT ALLOWS THE USER TO CHANGE THE GAP ELEMENT TYPE OF GAP ELEMENTS DEFINED IN THE MODEL FOR PARTICULAR LOAD CASES. THIS DATA CAN BE REPEATED AS NECESSARY TO SELECT ALL THE DESIRED GAP ELEMENTS.
(7-10)	ENTER THE LOAD CASE TO WHICH THESE OVERRIDES APPLY.
(12-14)	SELECT EITHER INCLUDE ('INC') TO SPECIFY THOSE GAP ELEMENT MEMBERS THAT ARE TO BE MODIFIED FOR THIS LOAD CASE OR EXCLUDE ('EXC') TO SPECIFY THAT ALL GAP MEMBERS EXCEPT THOSE SPECIFIED ARE TO BE MODIFIED. NOTE THAT THE EXCLUDES AND INCLUDES CANNOT BE MIXED FOR ANY ONE LOAD CASE.
(16-17)	ENTER THE GAP ELEMENT TYPE FROM THE FOLLOWING CHOICES: (Default: 'CO')
(19-21)	OVERRIDE TYPE: 'ST' - STANDARD SACS ELEMENT 'CO' - COMPRESSION ONLY. 'TO' - TENSION ONLY. 'NL' - NO LOAD. 'FD' - FORCE VERSUS DEFLECTION ELEMENT (SEE 'F-DEL' DATA). 'RP' - REPEAT PREVIOUS GAP ELEMENT FORCE-DEFLECTION.
(23-26)	ENTER THE START JOINT OF THE FIRST GAP ELEMENT.
(27-30)	ENTER THE END JOINT OF THE FIRST GAP ELEMENT.
(32-75)	ENTER THE REMAINING GAP ELEMENTS.

[illegible]

END OF INPUT

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	THIS LINE IS THE LAST LINE IN THE 'GAP' INPUT FILE.
GENERAL	THE 'END' LINE TERMINATES THE DATA READ BY THE 'GAP' PROGRAM. THIS LINE IS OPTIONAL.

LINE LABEL	REMAINDER OF THIS INPUT LINE LEFT BLANK
END	
1-- 3	4-----80

FORCE VERSUS DEFLECTION CURVE DEFINITION

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS REQUIRED FOR EACH NONLINEAR GAP ELEMENT THAT HAS A 'FD' DESIGNATION. THIS DATA FOLLOWS DIRECTLY AFTER THE 'GAPELM' LINE DEFINING THE GAP ELEMENT TO WHICH THIS FORCE VERSUS DEFLECTION RELATIONSHIP APPLIES.
(9-26)	ENTER THE FIRST FORCE-DEFLECTION POINT TO DEFINE THIS CURVE.
(27-80)	ENTER THE SECOND, THIRD AND FOURTH POINTS AS REQUIRED. THESE POINTS MUST BE ENTERED IN ORDER OF INCREASING DEFLECTIONS. FOR DEFLECTIONS OUTSIDE THE RANGE OF THE TABLE VALUES, THE PROGRAM USES THE FIRST OR LAST FORCE VALUE IN THE TABLE AS APPROPRIATE. REPEAT THIS LINE AS OFTEN AS NECESSARY TO DEFINE THE FORCE VERSUS DEFLECTION CURVE IN AS MUCH DETAIL AS DESIRED. DO NOT USE BLANKS FOR ZEROS.

LINE LABEL	FIRST POINT		SECOND POINT		THIRD POINT		FOURTH POINT	
	FORCE	DEFL.	FORCE	DEFL.	FORCE	DEFL.	FORCE	DEFL.
F-DEL								
1-- 5	9<--17	18<--26	27<--35	36<--44	45<--53	54<--62	63<--71	72<--80
DEFAULT								
ENGLISH	KIP	IN	KIP	IN	KIP	IN	KIP	IN
METRIC (KN)	KN	CM	KN	CM	KN	CM	KN	CM
METRIC (KG)	KG	CM	KG	CM	KG	CM	KG	CM

GAP ELEMENT INPUT

COLUMNS	COMMENTARY
GENERAL	THIS INPUT LINE IS REQUIRED FOR EACH NONLINEAR GAP ELEMENT. IT PROVIDES THE ELEMENT DEFINITION TO DUMMY LOAD CASE CONNECTION. THIS DATA IS NOT REQUIRED FOR GAP ELEMENTS DEFINED IN THE SACS IV INPUT FILE.
(8-11)	ENTER THE FIRST JOINT OF THIS GAP ELEMENT.
(13-16)	ENTER THE SECOND JOINT OF THIS GAP ELEMENT.
(18-21)	ENTER THE DUMMY LOAD CASE NAME ASSOCIATED WITH THIS ELEMENT. THIS LOAD CASE WOULD HAVE A LOAD APPLIED ONLY TO THIS ELEMENT.
(24-25)	ENTER THE GAP ELEMENT TYPE FROM THE FOLLOWING CHOICES: 'CO' - COMPRESSION ONLY. 'TO' - TENSION ONLY. 'NL' - NO LOAD. 'FD' - FORCE VERSUS DEFLECTION ELEMENT (SEE 'F-DEL' DATA). 'RP' - REPEAT PREVIOUS GAP ELEMENT FORCE-DEFLECTION. 'TC' - COMPRESSION ONLY TIE-DOWN. 'TT' - TENSION ONLY TIE-DOWN. 'TD' - TIE-DOWN.

LINE LABEL	GAP ELEMENT		DUMMY LOAD CASE NAME	ELEMENT TYPE	LEAVE BLANK
	FIRST JOINT	SECOND JOINT			
GAPELM					
1-- 6	8-->11	13-->16	18-->21	24--25	26-----80
DEFAULT				'CO'	

'GAP' OPTIONS INPUT

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS REQUIRED IN ANY NONLINEAR GAP ELEMENT ANALYSIS. IT SPECIFIES OVERALL ANALYSIS PARAMETERS.
(7-10)	ENTER THE NUMBER OF REAL LOAD CASES IN THE SACS IV DATA. THE REAL LOAD CASES ARE THOSE THAT REPRESENT ACTUAL LOADING ON THE STRUCTURE AS OPPOSED TO THE DUMMY LOAD CASES THAT ARE USED TO SIMULATE THE GAP ELEMENTS.
(11-14)	ENTER THE NUMBER OF OUTPUT LOAD CASES. THESE ARE THE LOAD CASES THAT REPRESENT THE ACTUAL LOADING CONDITIONS TO BE ANALYZED AND IN GENERAL THESE WILL BE COMBINATIONS OF THE REAL LOAD CASES IN THE SACS IV DATA.
(15-18)	THIS OPTION IS PRIMARILY USED FOR DEBUGGING ANY PROBLEMS THAT MAY EXIST. IT WILL PRODUCE A GREAT DEAL OF OUTPUT THAT WILL NOT BE USEFUL FOR THE NORMAL CASE.
(19-20)	ENTER 'TW' IF THE AUTOMATIC 'TOW' ANALYSIS IS BEING USED. THIS OPTION REQUIRES THAT THE FIRST ACTUAL LOAD CASE IS A 'DEAD' LOAD CASE. THE FOLLOWING LOAD CASES ARE MOTION LOAD CASES WITHOUT GRAVITY. IF GRAVITY IS INCLUDED IN THE MOTION LOAD CASES, THEN ENTER 'TG' FOR THIS OPTION. THE TIE DOWNS ARE SPECIFIED IN THE SACS IV DATA AS NULL ELEMENTS.

COLUMNS	COMMENTARY
(21-22)	ENTER THE SYSTEM OF UNITS USED. THIS SYSTEM WILL NORMALLY BE THE SAME AS USED IN THE SACS IV ANALYSIS, BUT IS NOT REQUIRED TO BE SO. SELECT FROM ONE OF THE FOLLOWING: 'EN' - ENGLISH UNITS. 'MN' - METRIC UNITS WITH KILONEWTONS AS THE FORCE UNITS. 'ME' - METRIC UNITS WITH KILOGRAMS AS THE FORCE UNITS.
(23-26)	ENTER THE NUMBER OF ITERATIONS ALLOWED IN THE NONLINEAR SOLUTION.
(27-34)	ENTER THE CONVERGENCE TOLERANCE REQUIRED. THIS TOLERANCE SHOULD BE SMALL SINCE THE GAP ELEMENTS TYPICALLY ARE SHORT AND RELATIVELY STIFF.
(35-44)	ENTER THE FRICTION COEFFICIENTS FOR THE FRICTION ELEMENTS CREATED IN THE SACS IV INPUT FILE. THIS DATA CAN BE OVERRIDDEN FOR INDIVIDUAL ELEMENTS ON THE 'GAPELM' LINE.
(45-47)	ENTER 'PFD' OR 'PFG' TO PLOT FORCE VERSUS DEFLECTION FOR GAP ELEMENTS. 'PFD' WILL PLOT WITHOUT GRID LINES; 'PFG' WILL PLOT WITH GRID LINES. WITH BLANK INPUT NO PLOTS WILL BE PRODUCED.
(77-78)	LEAVE BLANK FOR THE STANDARD SOLVER (DEFAULT). ENTER 'A1' FOR ADVANCED MATRIX INVERSION,OR ENTER 'A2' FOR ADVANCED LINEAR SOLVER. ENTER 'AD' FOR THE DIRECT SOLVER. THE DIRECT SOLVER IS ONLY APPLICABLE TO MODELS WITH TENSION-ONLY, COMPRESSION-ONLY, AND/OR NO-LOAD ELEMENTS WITH ZERO INITIAL GAP VALUES. THE PROGRAM AUTOMATICALLY SWITCHES TO 'A2' IF IT CANNOT USE DIRECT SOLVER FOR ANY REASON.
(79-80)	ENTER 'NE' TO CONTINUE THE GAP ANALYSIS ANALYSIS IN THE CASE NON-CONVERGENCE WITH A WARNING MESSAGE INDICATING THE NON-CONVERGENCE. LEAVE BLANK FOR DEFAULT WORKFLOW (STOP WITH AN ERROR MESSAGE INDICATING THE NON-CONVERGENCE) (NOTE) THE RESULTS MUST BE DOUBLE-CHECKED IF THE NON-CONVERGENCE CONTINUE OPTION IS USED.

LINE LABEL	NUMBER OF REAL LOAD CASES	NUMBER OF OUTPUT LOAD CASES	DIAGNOSTIC OUTPUT PRINT OPTION	AUTOMATIC TOW ANALYSIS OPTION	UNITS OPTION	MAXIMUM NUMBER OF ITERATIONS	CONVERGENCE TOLERANCE	FRICTION COEFFICIENTS		FORCE DEFLECTION PLOT	LEAVE BLANK	ADAVNCED SOLVER	CONTINUE OPTION FOR NON-CONVERGENCE?
								LOCAL Y	LOCAL Z				
GAPOPT													
1-- 6	7-->10	11-->14	15-->18	19--20	21--22	23-->26	27<--34	35<--39	40<--44	45--47	48--76	77-78	79-80
DEFAULT					'EN'	1000	0.00001	0.5	0.5			STANDARD SOLVER	STOP IN THE CASE OF NON-CONVERGENCE

'GAP' LOAD COMBINATION INPUT

COLUMNS	COMMENTARY
LOCATION	LOAD COMBINATIONS FOLLOW THE BASIC LOAD CONDITION DATA.
GENERAL	THIS LINE ENABLES THE USER TO GENERATE NEW LOAD CONDITIONS, EACH DEFINED AS A LINEAR COMBINATION OF FROM ONE TO FORTY EIGHT BASIC AND/OR OTHER COMBINED LOAD CONDITIONS FOR THIS ANALYSIS.
(1- 5)	ENTER 'LCOMB' ON ALL LINES DEFINING COMBINATIONS. A HEADER WITH 'LCOMB' ONLY MUST PRECEDE ANY LOAD COMBINATION DATA.
(7-10)	ENTER THE NAME FOR THE LOAD COMBINATION BEING DEFINED.
(12-15)	ENTER THE NAME OF THE LOAD CASE OR COMBINATION TO BE USED AS THE FIRST LOAD COMPONENT DEFINING THIS COMBINATION. THE LOAD CONDITIONS BEING COMBINED MAY BE ENTERED IN RANDOM ORDER.
(16-21)	ENTER THE FRACTION OF THE FIRST LOAD CASE TO BE INCLUDED IN THIS COMBINATION.
(22-71)	REPEAT AS NECESSARY FOR THE REMAINING COMPONENTS MAKING UP THIS COMBINATION.
(72-76)	ENTER THE ALLOWABLE STRESS MODIFIER FOR THIS LOAD CASE.
NOTE:	THIS LINE MAY BE REPEATED TO ENTER A TOTAL OF FORTY EIGHT LOAD COMPONENTS FOR EACH COMBINATION. EACH ADDITIONAL 'LCOMB' LINES MUST HAVE THE LOAD COMBINATION NAME SPECIFIED IN COLUMNS 7-10.

[illegible]

'GAP' ANALYSIS LOAD CASE SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY THE LOAD CASES IN THE SACS IV INPUT FILE THAT ARE TO BE USED IN THE 'GAP' PROGRAM. THIS LINE CAN BE REPEATED AS OFTEN AS NECESSARY TO SELECT ANY OR ALL OF THE LOAD CASES.
(17-75)	ENTER THE LOAD CASE IDENTIFIERS FOR ALL LOAD CASES TO BE INCLUDED FOR 'GAP' ANALYSIS. THE LOAD CASES CAN BE IN ANY ORDER.

LINE LABEL	LOAD CASE SELECTION											
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH
LCSEL												
1-- 5	17-->20	22-->25	27-->30	32-->35	37-->40	42-->45	47-->50	52-->55	57-->60	62-->65	67-->70	72-->75